

## Form I – 3A

## INTERN'S DAILY DIARY

(To be filled by the Intern- Please ensure to upload duly filled set of forms at end of four weeks to the provided folder)

## Intern's Information

Intern's Name

D.N Gamage

Student ID

IT23481244

Internship Title

Software Engineer (Intern)

Specialisation

Information Technology

Supervisor  
Name

Bandara K.M.V.T

## Internship Information

## Training Information For the Week (to be filled by the intern)

| DATE       | DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., SKETCHES AND DIMENSIONS TO BE GIVEN WHEREVER POSSIBLE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 08/09/2025 | <p>Today was my first day as a software engineering intern at Vihi it solutions. The day began with an orientation session where I officially signed my internship agreement and was introduced to the company's policies, workflows, and project management structure.</p> <p>Following the training, I was introduced to the software development team and assigned the role of Project Team. My supervisor provided an overview of the project, its objectives, and the expected deliverables. I also took some time to familiarize myself with the team members I will be coordinating throughout the internship.</p> <p>Goals:</p> <ul style="list-style-type: none"> <li>- Sign internship agreement</li> <li>- Complete orientation session</li> <li>- Learn company policies and workflows</li> <li>- Meet the development team</li> <li>- Receive project assignment</li> <li>- Get introduced to team members</li> <li>- Set up company email and communication tools</li> </ul> |

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|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 09/09/2025 | <p>The second day focused on setting up my development environment and getting hands-on with the technical tools and systems used by team. I configured my workstation with all the necessary software development tools. This was a crucial step to ensure I could actively contribute to the project in the coming days.</p> <p>I cloned the company web's project repository and successfully set up the local development environments. Running the project locally for the first time gave me a sense of accomplishment and helped me understand the project structure better. I also spent time reviewing the team's coding standards and best practices to align my work with the company's quality expectations.</p> <p><b>Goals:</b></p> <ul style="list-style-type: none"> <li>- Set up development workstation with all required tools</li> <li>- Configure version control and understand the Git workflow</li> <li>- Clone and run the project locally</li> <li>- Familiarize with coding standards and development practices</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 10/09/2025 | <p>Today was dedicated to deep diving into the project codebase and understanding its architecture. A company front-end developer, Mr. Dumindu spoke to us about technologies and gave reasons why they are now essential in making modern web applications that are responsive and scalable. I spent considerable time reading through the project documentation to grasp the overall system design and business logic. Exploring the codebase structure helped me identify where different components, services, and utilities are organized</p> <p>I studied the technology stack in detail - React for the frontend, Next.js for server-side rendering, and Material-UI for the component library. Running and testing existing features gave me practical insights into how the application works from a user's perspective. I documented several questions that arose during my exploration to discuss with the team in our next meeting.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 11/09/2025 | <p>The fourth day introduced me to the team's workflow and collaboration practices. I attended my first daily standup meeting, where I observed how the team communicates progress, blockers, and plans for the day. This gave me valuable insight into the agile methodology practiced at Vihi IT Solutions.</p> <p>I completed several internal training modules covering the company's development processes and best practices. Learning the project management tools was essential for tracking tasks and understanding sprint progress. Shadowing a senior developer was particularly enlightening - I observed their problem-solving approach, code review techniques, and how they handle complex features. I also learned about the pull request process and deployment workflow, which will be crucial for my contributions.</p> <p><b>Goals:</b></p> <ul style="list-style-type: none"> <li>- Understand the team's daily workflow and communication</li> <li>- Learn project management and collaboration tools</li> <li>- Observe senior developer's work practices</li> <li>- Understand code review and deployment processes</li> </ul> <p><b>Tasks:</b></p> <ul style="list-style-type: none"> <li>- Attend team standup meeting</li> <li>- Complete internal training modules</li> <li>- Learn project management tools (Jira/Trello/Azure DevOps)</li> <li>- Shadow senior developer</li> <li>- Review pull request process</li> <li>- Learn deployment workflow</li> <li>- Set up development branch</li> </ul> |

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|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12/09/2025 | <p>Today as an exciting milestone - receiving my first actual development task! My supervisor, K.M.V.T Bandara assigned me a manageable task to help me get started with hands-on coding. I learned the important skill of breaking down a larger task into smaller, actionable subtasks, making the work more approachable and easier to track.</p> <p>Before diving into coding, I discussed my proposed approach with my mentor, which helped me validate my understanding and receive valuable guidance. I started working on a simple bug fix or UI update, applying the coding standards and practices I had learned over the past few days. The weekly team meeting provided a great opportunity to see the bigger picture of what the team is working on.</p> <p>As the first week concluded, I took time to document my learnings, reflecting on the tremendous amount of knowledge I had absorbed and the progress I had made from orientation to active development.</p> <p>Goals:</p> <ul style="list-style-type: none"> <li>- Receive and understand first development task</li> <li>- Learn task breakdown and planning</li> <li>- Get mentor guidance before implementation</li> <li>- Start contributing code to the project</li> <li>- Reflect on first week's learnings</li> </ul> <p>Tasks:</p> <ul style="list-style-type: none"> <li>- Receive first task assignment</li> <li>- Break down task into subtasks</li> <li>- Discuss approach with mentor</li> <li>- Start working on simple bug fix or UI update</li> <li>- Weekly team meeting</li> <li>- Document learnings from the week</li> </ul> |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## SUPERVISOR COMMENTS FOR THE WEEK

*Gamage has demonstrated a highly positive and proactive start to her internship. She successfully completed the onboarding process, set up her local development environment, and quickly familiarized herself with our Next.js and React technology stack. Her active participation in daily standups and ability to properly break down her first assigned development task shows excellent foundational skills and adaptability.*

Supervisor's  
Signature



Date 9/19/2025

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Specialisation

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Supervisor  
Name

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|------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 15/09/2025 | Academic Holiday (Mid-Exam)                                                                                                             |
| 16/09/2025 | Academic Leave (Mid-Term)                                                                                                               |
| 17/09/2025 | Academic Leave (Mid-Term)                                                                                                               |
| 18/09/2025 | Academic Leave (Mid-Term)                                                                                                               |

19/09/2025

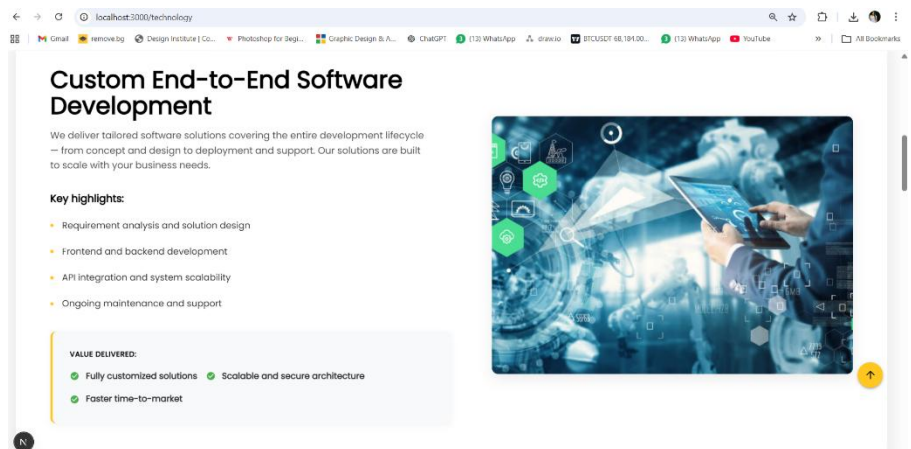
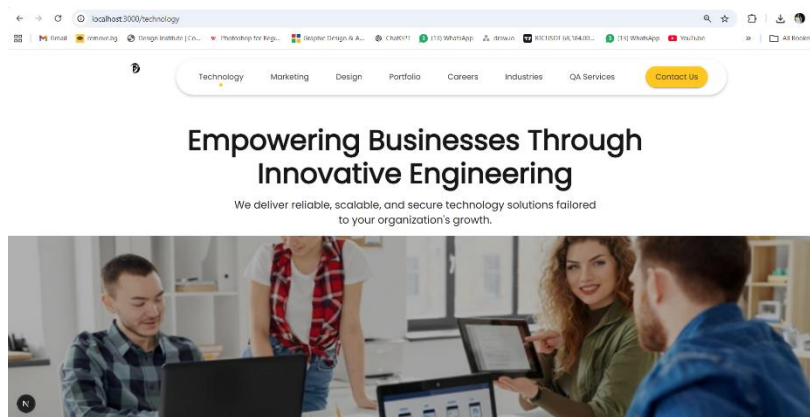
The day after the initial, I continued development on the Technology page that create another branch, focusing on UI enhancement and refinement rather than simply completing the initial implementation. My primary focus was on redesigning the user interface layout to achieve optimal visual hierarchy and component alignment, ensuring consistency with the company's overall design system.

I leveraged Material-UI components including Cards, Typography, Grid, and Icons to structure and present technology-related information in a clean, organized manner. The implementation involved strategic positioning of elements, proper spacing configurations, and maintaining visual consistency across all sections. Special attention was given to responsive design principles, implementing fluid layouts and breakpoint-specific styling to ensure seamless functionality across various viewport sizes and devices.

This hands-on experience deepened my understanding of the company's front-end development standards, design patterns, and workflow practices, marking my transition from foundational learning to practical application of professional development principles. As the first week concluded, I documented my learnings and reflected on the substantial progress from orientation to active feature development.

Goals:

- Implement and enhance Technology page UI
- Apply Material-UI component library effectively
- Ensure responsive design across all breakpoints
- Maintain consistency with company design system
- Understand frontend development workflow



## SUPERVISOR COMMENTS FOR THE WEEK

*Although Gamage was on academic leave for the majority of the week, her return on Friday was highly productive. She successfully applied Material-UI components to enhance the Technology page's layout, ensuring it met our responsive design and visual consistency standards. Her ability to seamlessly transition from foundational learning into practical feature development in such a short window is commendable.*

Supervisor's  
Signature



Date

9/26/2025

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|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 22/09/2025 | <p>Today company decide to get a project meeting where the team reviewed current development initiatives, progress milestones, and upcoming sprint objectives. The discussion covered various aspects of the company's web development roadmap, including technical requirements, feature prioritization, and delivery timelines. I actively documented key technical decisions and architectural considerations, gaining valuable insights into the software development lifecycle and agile project management practices employed at Vihi IT Solutions.</p> <p>The meeting provided crucial context regarding team collaboration workflows, code review procedures, and quality assurance standards. Understanding the broader project landscape and how individual tasks contribute to overall business objectives deepened my appreciation for strategic planning in software development.</p> <p>Following the meeting, I returned to frontend development tasks, conducting a systematic review and refinement of existing UI components. My focus centered on identifying and resolving minor but impactful issues including layout alignment inconsistencies, responsive design breakpoints, and visual design system adherence. I applied user-centric design principles while implementing fixes, leveraging feedback from code reviews and design specifications. This iterative refinement process reinforced the critical importance of attention to detail and the cumulative impact of small improvements on overall application quality and user experience.</p> <p>Goals:</p> <ul style="list-style-type: none"> <li>- Understand project workflow and development processes</li> <li>- Gain insight into team dynamics and collaboration patterns</li> <li>- Identify and resolve frontend UI/UX issues</li> <li>- Apply feedback-driven iterative improvements</li> <li>- Strengthen attention to detail in production code</li> </ul> |

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|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 23/09/2025 | <p>Today marked the successful completion of the Technology Page as well as Marketing Page development. I conducted a comprehensive final review, implementing critical adjustments to ensure optimal component alignment, responsive behavior across all viewport breakpoints, and strict adherence to UI consistency standards. Leveraging Material-UI's component library within the Next.js framework, I refined spacing configurations, grid layouts, and typography hierarchies to achieve a polished, production-ready implementation.</p> <p>Following the completion and review of the Technology Page, I received my next frontend assignment: development of the company website's Design Page. I participated in a detailed briefing session with the senior frontend developer, who outlined the page's objectives, content architecture, and design specifications. The discussion covered user journey considerations, information hierarchy, and visual design expectations aligned with the company's brand guidelines.</p> <p>With a clear understanding of the requirements, I initiated the planning phase for the Design Page. I analyzed layout requirements, sketched component structures, and developed a preliminary implementation strategy. This architectural planning phase involved identifying reusable components from the existing codebase, determining state management needs, and mapping out the responsive design approach before proceeding to the actual implementation phase.</p> <p>Goals:</p> <ul style="list-style-type: none"> <li>- Complete and finalize Technology Page development</li> <li>- Ensure production-ready code quality and responsiveness</li> <li>- Understand Design Page requirements and objectives</li> <li>- Plan component architecture and implementation strategy</li> <li>- Apply lessons learned from Technology Page development</li> </ul> |
| 24/09/2025 | <p>Today I implemented a complete feature using multiple components working together. I studied the existing codebase design patterns to maintain consistency and followed the established coding conventions.</p> <p>I worked with React hooks like useState and useEffect for state management, learning when to lift state to parent components. I added error handling to manage edge cases and provide clear error messages. I added code comments where needed for readability and conducted a self-review to catch any issues.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 25/09/2025 | <p>The supervisor, K.M.V.T Bandara organized a meeting. Participating in the code review session was incredibly educational - I learned how to explain my implementation decisions and understand different perspectives on code quality. The reviewers provided thoughtful feedback that helped me see aspects of my code I hadn't considered.</p> <p>Addressing the reviewer's feedback was more than just making changes - it was about understanding the 'why' behind each suggestion. I refactored sections of code to be more efficient, readable, and maintainable. Some suggestions involved design pattern improvements, while others focused on edge cases I hadn't handled. Updating my tests to cover these new scenarios ensured the code remained reliable. I felt more confident knowing the code had been thoroughly reviewed and improved.</p> <p>Goals:</p> <ul style="list-style-type: none"> <li>- Successfully navigate code review process</li> <li>- Learn from reviewer feedback</li> <li>- Improve code quality through refactoring</li> <li>- Update tests to match changes</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 26/09/2025 | <p>I focused on quality assurance and testing. I performed manual testing across different user flows and scenarios, identifying several UI issues and unexpected behaviors that automated tests had not detected. I systematically tested edge cases and documented all findings for future reference.</p> <p>Based on the identified issues, I implemented bug fixes and created additional test cases to prevent similar problems in the future. My weekly progress review with my supervisor covered the week's accomplishments, challenges faced, and areas for improvement. Together we planned next week's tasks and set clear objectives for continued skill development.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |



## SUPERVISOR COMMENTS FOR THE WEEK

*Gamage made significant contributions this week by finalizing both the Technology and Marketing pages to a production-ready standard. She handled the transition to planning the Design page well and showed a strong understanding of state management using React hooks. Furthermore, her receptiveness to feedback during code reviews and her initiative in conducting thorough manual QA testing demonstrate a growing maturity in writing maintainable, high-quality code.*

Supervisor's  
Signature



Date

10/3/2025

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|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 29/09/2025 | <p>Today, I participated in a progress review meeting with Mr. Dumindu to present the current status of the company website development. During the meeting, I demonstrated all developed pages including the Technology Page and other completed sections, highlighting the implementations, UI enhancements, and responsive design features achieved over the past days.</p> <p>Mr. Dumindu conducted a comprehensive review of the developed pages, identifying several issues related to layout consistency, responsive behavior, and cross-browser compatibility. I documented all identified issues and implemented the necessary fixes, including alignment adjustments, breakpoint refinements, and UI consistency improvements. Following the fixes, I performed thorough testing across different browsers and devices to ensure all issues were resolved. The meeting concluded with planning next week's development tasks and setting clear objectives for continued progress.</p> <p>Goals:</p> <ul style="list-style-type: none"> <li>- Present web development progress to Mr. Dumindu</li> <li>- Demonstrate completed pages and features</li> <li>- Address identified issues and implement fixes</li> <li>- Conduct comprehensive testing and validation</li> </ul> |
| 30/09/2025 | <p>Today marked the beginning of my assigned task at Vihi IT Solutions – designing the physical network floor plans for the company's new three-story office building. This task was specifically assigned to me by my supervisor, Mr. K.M.V.T. Bandara, who entrusted me with sole responsibility for the floor plan design component.</p> <p>I began the day by conducting a thorough requirements analysis. I carefully studied the company's background, understanding that company is a growing software development and IT services company with 35 employees operating across development, administrative, and executive teams. I reviewed the full project scope, which</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

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|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            | <p>required designing network infrastructure for all three floors of the office building.</p> <p>Key activities carried out today:</p> <ul style="list-style-type: none"> <li>• Studied the company profile, including its core services (custom software development, mobile app development, IT consulting, and QA testing) and team composition to understand connectivity requirements for each department.</li> <li>• Analysed the building dimensions: 18m × 12m footprint across three floors (Ground, First, and Second), identifying the key functional areas for each floor.</li> <li>• Drafted an initial room-by-room list of network requirements.</li> <li>• Selected draw.io as the design tool for creating the floor plans, as it supports professional-grade network and architectural diagrams with a comprehensive symbol library.</li> <li>• Identified the required network symbols to be used: Data Outlets (☒), Access Points (orange squares), Equipment Racks (double-cross squares), Servers (purple rectangles), Printers, and Displays/Wall Displays (yellow rectangles).</li> </ul> <div style="text-align: center;">  Data Outlet<br/>  Printer<br/>  Access Point<br/>  Displays (Wall Displays/TV/Phone)<br/>  Server<br/>  Equipment Rack         </div> <p>By the end of the day I had a clear and structured understanding of what each floor required in terms of network components and was fully prepared to begin the design process on the following day.</p> |
| 01/10/2025 | <p>Today I focused entirely on designing the physical network floor plans for all three floors of the Vihi IT Solutions office building using draw.io. Based on yesterday's requirements analysis, I proceeded to create detailed and scaled floor plan diagrams, ensuring each room had the appropriate network components positioned accurately.</p> <div style="display: flex; justify-content: space-between;"> <div style="width: 48%;"> <p>Ground Floor Design:</p> <ul style="list-style-type: none"> <li>• Reception Area</li> <li>• Server Room</li> <li>• General Office</li> <li>• Meeting Room A</li> <li>• Meeting Room B</li> <li>• Admin Office</li> <li>• Storage</li> </ul> </div> <div style="width: 48%;"> <p>First Floor Design:</p> <ul style="list-style-type: none"> <li>• Development Team Area (largest space)</li> <li>• Team Lead Room</li> <li>• Conference Room</li> <li>• Testing Lab</li> <li>• IT Equipment Room</li> <li>• PM Office</li> <li>• Break Room</li> </ul> </div> </div> <p>Second Floor Design:</p> <ul style="list-style-type: none"> <li>• Executive Suite (CEO, CTO, Finance, HR)</li> <li>• Board Room</li> <li>• Client Meeting Room</li> <li>• Marketing Department</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

- Sales Room
- Accounts

By end of day, all three floor plan diagrams were completed and saved. I presented the draft designs to Mr. Bandara for initial review.



**Ground Floor**



**First Floor**



**Second Floor**

02/10/2025

Today I focused on developing the logical network topology diagram and implementing the Cisco Packet Tracer simulation for all three floors.

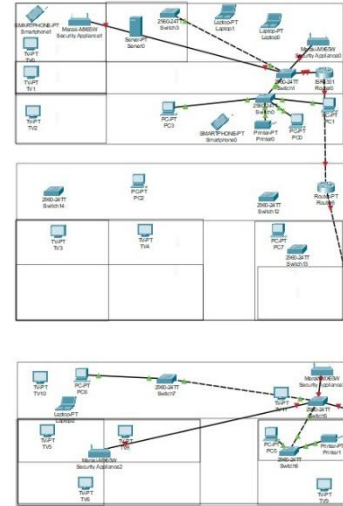
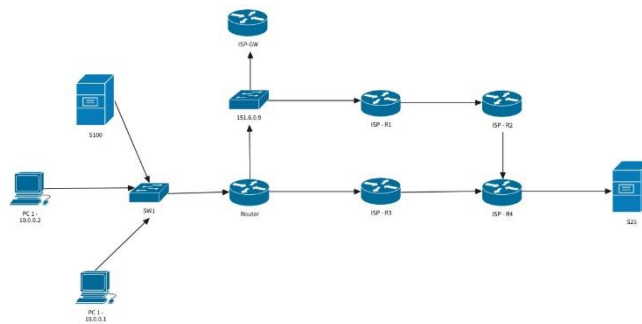
Network Topology Diagram:

- Designed a hierarchical topology: PCs (10.0.0.1, 10.0.0.2) and server S100 connecting to Switch SW1, then to a central Router linked to ISP infrastructure (ISP-R3, ISP-R4) reaching server S25.
- Added a redundant ISP path via Router 151.6.0.9 → ISP-R1 → ISP-R2 → ISP-R4 → ISP-GW for business continuity.
- Applied IP addressing: internal LAN 10.0.0.0/24 and ISP-facing 151.6.x.x range.

Cisco Packet Tracer Simulation:

- Ground Floor: Configured switches, routers, security appliances, server, laptops, PCs, printers, and TVs with wired and wireless links.
- First Floor: Set up switches and routers representing the development team's connectivity with PCs and TVs.
- Second Floor: Configured switches, security appliances, router, PCs, laptops, TVs, and a printer for the executive floor.
- Resolved a routing issue on Floor 2 where the router was not correctly linked to the upstream switch.

By end of day, the topology diagram and Packet Tracer simulation were fully functional and ready for documentation.



03/10/2025

On the final day of this task, I completed the cost estimation and the project documentation report.

#### Cost Estimation:

- Prepared a Bill of Materials (BOM) covering active components (switches, routers, 8 access points, firewall, servers), passive components (Cat6 UTP cabling, patch panels, wall plates, cable trays), and end devices (printers, IP phones).
- Verified all cable runs were within the 802.3 maximum of 90m given the 18m × 12m floor dimensions.
- Included installation costs, 10% contingency, and taxes, with the breakdown organized per floor.

#### Project Documentation Report:

- Compiled the full report covering company background, requirements analysis, logical and physical network design, Packet Tracer simulation, cost estimation, and scalability recommendations.

#### Final Review:

I submitted all deliverables to Mr. K.M.V.T Bandara. He commended the floor plan accuracy and suggested adding VLAN segmentation details to strengthen the security architecture documentation, which I noted for future enhancement.

This four-day task was a valuable experience, allowing me to apply networking theory in a real-world design scenario and develop skills in physical network design, Packet Tracer simulation, and technical documentation.

## SUPERVISOR COMMENTS FOR THE WEEK

*Gamage demonstrated excellent versatility this week. After effectively resolving frontend UI and cross-browser issues with Mr. Dumindu, she successfully pivoted to a completely different infrastructure task. She independently designed highly accurate physical network floor plans and logical topologies for the new three-story office building using draw.io and Cisco Packet Tracer. Her comprehensive final documentation, which included a detailed cost estimation and BOM, was well-structured, professional, and exceeded expectations.*

Supervisor's  
Signature

*Vihira*

Date

10/10/2025

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|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 06/10/2025 | Holiday(Poya Holiday)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 07/10/2025 | <p>On the first day of developing the Gym Management System for Mega Power Gym, I was officially assigned the project by Mr. K.M.V.T Bandara. The primary objective of the day was to clearly understand the overall operational environment of the gym and identify the challenges faced due to the existing manual system.</p> <p>To begin with, I conducted detailed discussions with the gym owner and the front-desk staff to analyze the current procedures, including member registration, attendance tracking, payment handling, and package renewals. Through these conversations, I identified that most records were maintained manually using notebooks and simple spreadsheets. This approach resulted in several issues such as duplicate member entries, difficulty in tracking active and expired memberships, and occasional errors in payment calculations.</p> <p>Moreover, I had a conversation regarding the expectations from the system in terms of automation, accuracy, and user-friendliness. All the information collected was very well documented. The requirement gathering process not only helped me to understand the business domain strongly but also ensured that the system proposed would be capable of handling the real-world operational challenges of Mega Power Gym effectively.</p> |

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|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 08/10/2025 | <p>Today, I focused on setting up the technical foundation of the project.</p> <p>I created a GitHub repository named “Gym-Management” to manage version control. Then, I initialized a React application using Create React App for the frontend development. The project folder structure was organized by separating frontend and backend components to ensure maintainability and scalability.</p> <p>Additionally, I configured Visual Studio Code with essential extensions for React development, Git integration, and code formatting to create an efficient development environment.</p>                                                                                                              |
| 09/10/2025 | <p>Designed the database architecture for the Gym Management System based on the gathered requirements. Identified major entities such as Members, Staff, Attendance, Payments, Appointments, Equipment, Schedules, Training Sessions, Feedback, and Ratings.</p> <p>Defined attributes, data types, and constraints for each table while ensuring normalization to eliminate data redundancy. Established relationships using foreign keys to maintain referential integrity between tables.</p> <p>Skills Developed:</p> <ul style="list-style-type: none"> <li>• Database design and normalization</li> <li>• ER modeling and relational mapping</li> <li>• Analytical thinking in system design</li> </ul> |
| 10/10/2025 | <p>Today ,Implemented the database design by creating the setup_database.sql script with all necessary CREATE TABLE statements, primary keys, foreign keys, and indexes.</p> <p>Inserted sample data for testing purposes, including member registrations, staff roles, and attendance records. Verified the database structure with the gym owner to ensure it aligned with operational requirements, especially payment handling and status tracking.</p> <p>Skills Developed:</p> <ul style="list-style-type: none"> <li>• SQL scripting and database implementation</li> <li>• Query optimization using indexes</li> <li>• Requirement validation with stakeholders</li> </ul>                             |

## SUPERVISOR COMMENTS FOR THE WEEK

*Gamage had an excellent start to her second month by taking ownership of the new Gym Management System project. She demonstrated strong analytical and communication skills during the requirement-gathering phase with the client. Furthermore, her ability to successfully translate those business requirements into a well-normalized database schema and establish the foundational React and GitHub infrastructure is highly commendable.*

Supervisor's  
Signature



Date

10/17/2025



## Form I – 3A

## INTERN'S DAILY DIARY

(To be filled by the Intern- Please ensure to upload duly filled set of forms at end of four weeks to the provided folder)

## Intern's Information

Intern's Name

D.N Gamage

Student ID

IT23481244

Internship Title

Software Engineer (Intern)

Specialisation

Information Technology

Supervisor  
Name

Bandara K.M.V.T

## Internship Information

## Training Information For the Week (to be filled by the intern)

| DATE       | DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., SKETCHES AND DIMENSIONS TO BE GIVEN WHEREVER POSSIBLE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13/10/2025 | <p>With the database schema finalized, I began developing the backend infrastructure for the Mega Power Gym Management System. I selected Node.js with Express.js to handle concurrent requests efficiently, which is important for managing registrations, check-ins, and payments simultaneously.</p> <p>I created a Backend folder and initialized the project using npm. Required dependencies were installed, including Express.js for routing and middleware handling. Instead of writing traditional SQL queries, I chose Prisma ORM to provide type-safe database access and reduce security risks such as SQL injections.</p> <p>The backend was structured using the MVC (Model-View-Controller) architectural pattern. Inside the src folder, I organized separate directories for API routes, configuration files, and middleware. Feature-based modules were created for members, staff, attendance, payments, and appointments to ensure scalability and maintainability.</p> <p>Skills Developed:</p> <ul style="list-style-type: none"> <li>• Backend project initialization with Node.js</li> <li>• Express.js routing and middleware setup</li> <li>• MVC architecture implementation</li> <li>• Codebase structuring for scalability</li> </ul> |

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 14/10/2025 | <p>Today, I focused on configuring the database connection and integrating Prisma ORM with the backend. I created an .env file to securely store environment variables such as database credentials and JWT secret keys, ensuring sensitive data would not be exposed in the repository.</p> <p>Next, I translated the previously designed SQL schema into Prisma's schema definition language and configured the database connection settings.</p> <p>Finally, I tested the backend-database communication by writing a simple test script to establish a connection and execute a basic query. The successful connection confirmed that the backend infrastructure was properly integrated with the MySQL database.</p> <p>Skills Developed:</p> <ul style="list-style-type: none"> <li>• Prisma ORM configuration</li> <li>• Environment variable management</li> <li>• Secure backend setup practices</li> <li>• Database connectivity testing</li> </ul>                                                                                                                                                                                                                               |
| 15/10/2025 | <p>Today, I focused on creating comprehensive documentation to ensure long-term maintainability of the Gym Management System project.</p> <p>I developed a detailed SETUP_GUIDE.md that explains the full development environment setup process, including installation of Node.js, MySQL, and Git, configuration of environment variables, and running database migrations. I also included troubleshooting steps for common issues such as database connection errors and port conflicts.</p> <p>Additionally, I created a README.md file for the Backend folder explaining the overall architecture, folder structure, and guidelines for adding new API endpoints following RESTful conventions (e.g., GET for listing, POST for creating records). Naming conventions for routes, controllers, and database models were clearly documented to maintain consistency.</p> <p>Skills Developed:</p> <ul style="list-style-type: none"> <li>• Technical documentation writing</li> <li>• RESTful API standardization</li> <li>• Structured project onboarding preparation</li> </ul>                                                                                                       |
| 16/10/2025 | <p>On this day, I created a DATABASE_SETUP.md file explaining schema design decisions, table relationships, and migration processes. I documented the reasoning behind key design choices.</p> <p>I carefully reviewed the package.json files for both frontend and backend to ensure all dependencies were properly listed with versions. I added custom npm scripts for starting the server, running migrations, and executing routine tasks to standardize development operations.</p> <p>Furthermore, I enhanced the example.env file with detailed comments explaining each environment variable and its purpose, including placeholders for future integrations such as payment gateway API keys.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 17/10/2025 | <p>Today, Mr. K.M.V.T Bandara conducted a progress review meeting to evaluate the current status of the Gym Management System development.</p> <p>During the meeting, I presented the completed work, including requirement analysis, database design, backend setup, and project documentation. I explained the architectural decisions taken, the use of Node.js with Express, Prisma ORM integration, and the structured MVC implementation. The database schema and documentation files were also demonstrated.</p> <p>Constructive feedback was provided regarding code optimization, documentation clarity, and future feature planning. Suggestions were given to focus on API security, input validation, and proper testing in the upcoming development phase.</p> <p>The meeting helped ensure that the project is progressing according to expectations and aligned with academic and practical standards.</p> <p>Skills Developed:</p> <ul style="list-style-type: none"> <li>• Professional project presentation</li> <li>• Technical communication</li> <li>• Receiving and implementing constructive feedback</li> <li>• Project progress evaluation and planning</li> </ul> |

## SUPERVISOR COMMENTS FOR THE WEEK

*This week, Gamage showed strong backend development capabilities by effectively setting up the Node.js and Express infrastructure, adopting the MVC pattern, and integrating Prisma ORM. I am particularly impressed with her dedication to writing comprehensive technical documentation (SETUP\_GUIDE.md and DATABASE\_SETUP.md), which is a crucial professional habit for ensuring long-term project maintainability. She also presented her architectural decisions clearly during our progress review.*

**Supervisor's  
Signature**



**Date**

**10/24/2025**

## Form I – 3A

## INTERN'S DAILY DIARY

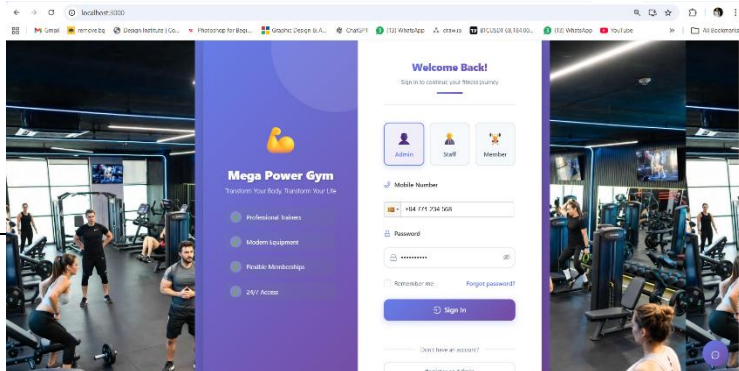
(To be filled by the Intern- Please ensure to upload duly filled set of forms at end of four weeks to the provided folder)

## Intern's Information

Intern's Name **D.N Gamage**Student ID **IT23481244**Internship Title **Software Engineer (Intern)**Specialisation **Information Technology**Supervisor Name **Bandara K.M.V.T**

## Internship Information

## Training Information For the Week (to be filled by the intern)

| DATE       | DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., SKETCHES AND DIMENSIONS TO BE GIVEN WHEREVER POSSIBLE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 20/10/2025 | Holiday                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 21/10/2025 | <p>Today, I began frontend development by creating the Admin Dashboard for the Mega Power Gym Management System. This dashboard serves as the central monitoring panel for management to view key metrics such as total active members, today's attendance, pending payments, and upcoming appointments.</p> <p>I designed a card-based layout in Dashboard.jsx to display important statistics clearly and efficiently. Each card included an icon, current value, and status indicator. Real-time data was fetched from the backend using React's useEffect hook, connected to a dedicated API endpoint for aggregated statistics.</p> <p>The dashboard was styled using CSS Grid to ensure a responsive and modern design. I also added a "Quick Actions" section with navigation links to common administrative tasks using React Router. Loading states and error handling were implemented to improve performance and user experience.</p>  |

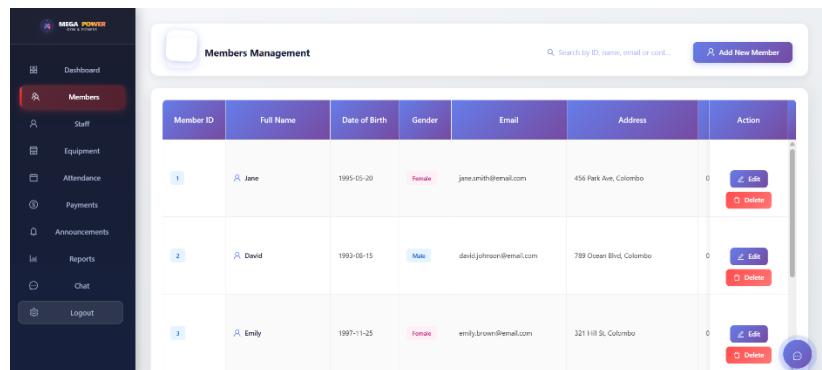


22/10/2025

Today, I focused on developing the Member Management module, a critical part of the Gym Management System. I created the Member.jsx registration form to capture personal, contact, emergency, membership, and health information. Real-time form validation was implemented to ensure data accuracy and prevent duplicates, including checks for email, phone number, and NIC uniqueness.

I also built the Membertable.jsx component to display registered members in a searchable, filterable table with key details like membership status, joining/expiry dates, and quick access to detailed profiles. Active and expired memberships were visually distinguished to help staff track renewals efficiently.

On the backend, I implemented API endpoints for member CRUD operations (POST, GET, PUT, DELETE), including soft delete functionality and duplicate detection.



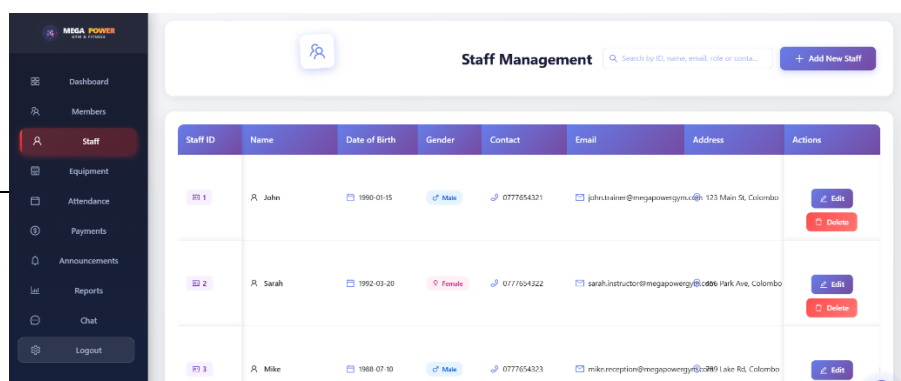
23/10/2025

Today, I developed the Staff Management module for Mega Power Gym, incorporating role-based access control (RBAC) to differentiate system access for Admins, Staff, and Trainers. I created the Staff.jsx registration form capturing personal details, roles, qualifications, and specializations, including trainer certifications and hourly rates.

On the backend, I implemented RBAC middleware to verify user roles via JWT tokens, ensuring that only authorized users could access specific endpoints. For example, only Admins can register staff or modify payments, while Trainers can manage training schedules and member progress.

The Stafftable.jsx component was built to list staff with roles, specializations, and status, along with action buttons for editing, password resets, or deactivating accounts. I also developed a StaffDashboard.jsx with role-specific views, showing tasks relevant to each staff type.

Security measures included password hashing with bcrypt, enforcing strong passwords, and account lockout after multiple failed login attempts.



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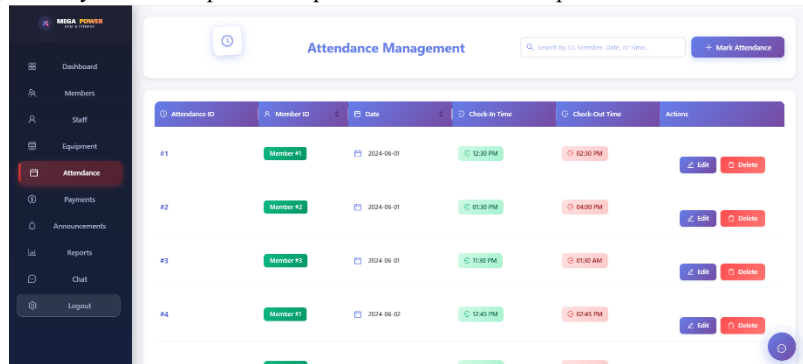
24/10/2025

Today, I focused on developing the Attendance module to automate Mega Power Gym's member check-in process. Previously, members signed a physical register, making it difficult to track attendance, generate reports, or identify inactive members.

I created the Attendance.jsx component with a simple search bar for quick member lookup, displaying photos for fast identification, and a one-click *Check In* button that automatically recorded date and time. The system included validation to prevent duplicate check-ins and automatically checked membership status, warning staff if a member's subscription had expired.

The Attendancetable.jsx component provided detailed attendance history with date range filtering, workout duration, and CSV export functionality. An attendance dashboard was also implemented to analyze peak hours, daily patterns, and low-attendance members, helping management optimize staffing and equipment usage. Database indexing on member\_id and check\_in\_datetime ensured fast queries for large datasets

At the end of the work, Mr. K.M.V.T Bandara conducted a progress review meeting to evaluate the current status of the Gym Management System development and provide feedback on completed modules.



### SUPERVISOR COMMENTS FOR THE WEEK

*Gamage made rapid and highly effective progress on the frontend this week, successfully delivering several core React components including the Admin Dashboard, Member Management, and Attendance modules. Her implementation of Role-Based Access Control (RBAC) via JWT middleware for the Staff Management module highlights her growing capability to handle complex, secure application logic alongside responsive UI design.*

Supervisor's  
Signature



Date

Click here to enter a date.

# Form I – 3A

## INTERN'S DAILY DIARY

(To be filled by the Intern- Please ensure to upload duly filled set of forms at end of four weeks to the provided folder)

### Intern's Information

Intern's Name **D.N Gamage**

Student ID **IT23481244**

Internship Title **Software Engineer (Intern)**

Specialisation **Information Technology**

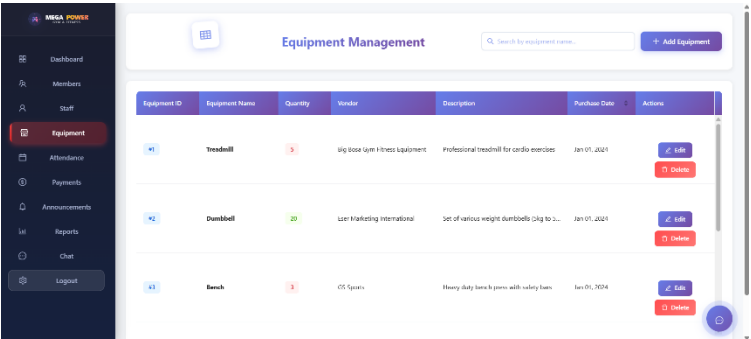
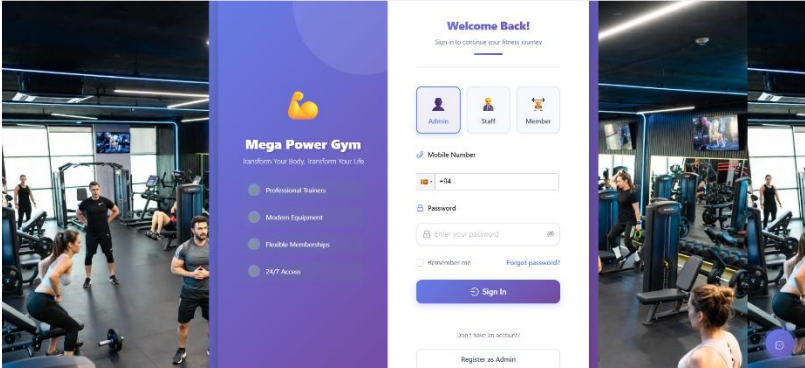
Supervisor Name **Bandara K.M.V.T**

### Internship Information

#### Training Information For the Week (to be filled by the intern)

| DATE       | DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., SKETCHES AND DIMENSIONS TO BE GIVEN WHEREVER POSSIBLE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 27/10/2025 | Lab Exam 2 (DSA)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 28/10/2025 | <p>Today, I developed the Appointment Booking module for Mega Power Gym, replacing the manual appointment book. I created the Appointment.jsx interface for staff and members to book sessions such as personal training, nutrition consultations, and fitness assessments. The form included service type, trainer selection, date/time, and special notes, with a calendar view showing available slots to prevent double bookings.</p> <p>On the backend, I implemented conflict detection logic to ensure trainers were not double-booked and added buffer times between appointments. The Appointmenttable.jsx component displayed upcoming appointments with color-coded statuses (confirmed, pending, reschedule) and filtering by date, trainer, and service type.</p> <p>Additionally, I built an appointment history feature to track past sessions, cancellations, and no-shows. Placeholder functions for email notifications were added for future confirmation and reminder automation.</p> |
| 29/10/2025 | <p>Today, Developed the Equipment Management system to track gym equipment and schedule maintenance. Created Equipment.jsx to register equipment details including category, purchase date, cost, warranty, and condition. Each item was assigned a unique ID.</p> <p>Implemented condition tracking with statuses (Excellent, Good, Needs Maintenance, Out of Service) that automatically flagged unavailable equipment in the scheduler. The Equipmenttable.jsx displayed sortable, filterable equipment lists with color-coded status badges.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                      |



|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            | <p>Added maintenance scheduling and history tracking with reminders, logging all maintenance actions, costs, and staff involved. A simple utilization report was also implemented to support future equipment management decisions.</p> <p><b>Skills Applied:</b><br/>Inventory tracking, React tables with filtering, status management, maintenance scheduling, report generation</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 30/10/2025 | <p>Today, I developed the login interface for the Mega Power Gym Management System. I created Login.jsx with a modern, responsive split-screen design featuring the gym logo and a motivational image alongside the login form.</p> <p>The form included email/username and password fields, a "Remember Me" checkbox, and a "Forgot Password?" link, with real-time validation and inline error messages. I implemented a password visibility toggle and role-based redirection to the Admin, Staff, or Member dashboard upon successful login.</p> <p>The interface was styled using Login.css with brand colors, responsive layouts, smooth transitions, and hover effects. Proper error handling was added to give informative yet secure feedback on login failures.</p> <p><b>Skills Applied:</b><br/>React form handling, responsive CSS, UI/UX design, client-side validation, React Router, local storage management</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 31/10/2025 | <p>Today, I focused on securing user credentials by implementing password hashing using bcrypt. Previously, passwords were stored in plain text, posing a serious security risk. I configured bcrypt with a cost factor of 10 to generate unique hashes with salt for every user, ensuring protection against rainbow table attacks.</p> <p>I updated member and staff registration to hash passwords before storage, enforced strong password validation (minimum 8 characters, uppercase, lowercase, number, special character) on both frontend and backend, and implemented secure login verification using bcrypt.compare. I also created a password update endpoint requiring the current password for changes, preventing unauthorized modifications.</p> <p>Thorough testing ensured no plain text passwords were stored or exposed, and login and password change processes were secure.</p> <p>Mr. K.M.V.T Bandara conducted a month-end progress review meeting to evaluate the current status of the Gym Management System development. During the meeting, I presented the completed modules, including frontend interfaces, backend functionalities, and security implementations. Feedback was provided on code quality, module integration, documentation, and upcoming development priorities.</p> <p><b>Skills Applied:</b><br/>bcrypt hashing, secure password validation, backend security, regex for password strength, safe password comparison.</p> |



## SUPERVISOR COMMENTS FOR THE WEEK

*Gamage concluded the month with outstanding work on the Appointment Booking and Equipment Management modules, implementing complex features like booking conflict detection and maintenance tracking. Most importantly, her initiative to proactively secure user credentials using bcrypt password hashing and strict validation rules demonstrates a mature, security-first approach to software engineering. She delivered a very strong month-end progress review.*

**Supervisor's  
Signature**



**Date** 11/7/2025

## Form I – 3A

## INTERN'S DAILY DIARY

(To be filled by the Intern- Please ensure to upload duly filled set of forms at end of four weeks to the provided folder)

## Intern's Information

Intern's Name

D.N Gamage

Student ID

IT23481244

Internship Title

Software Engineer (Intern)

Specialisation

Information Technology

Supervisor  
Name

Bandara K.M.V.T

## Internship Information

## Training Information For the Week (to be filled by the intern)

| DATE       | DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., SKETCHES AND DIMENSIONS TO BE GIVEN WHEREVER POSSIBLE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 03/11/2025 | <p>Today, I focused on strengthening system security by implementing JWT-based authentication and a secure password reset mechanism.</p> <p>I integrated JSON Web Tokens (JWT) to create a stateless authentication system. Upon successful login, a token containing user ID, email, role, and expiration time was generated and verified through custom middleware for protected routes. Role-based authorization was enforced to restrict access to sensitive endpoints. I also implemented token refresh logic and logout functionality with token invalidation for enhanced security.</p> <p>Additionally, I developed a secure password reset feature. The Forgotpw.jsx interface allowed users to request password resets via email. The backend generated secure, time-limited reset tokens using the crypto library and sent branded emails using Nodemailer. Security measures included rate limiting, expiration token (1 hour), prevention of email enumeration, and token invalidation after use.</p> <p>Comprehensive testing ensured secure handling of valid, expired, and invalid tokens.</p> <p>Skills Applied:</p> <p>JWT authentication, Express middleware, role-based authorization, secure token generation, Nodemailer integration, rate limiting, authentication security best practices.</p> |

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 04/11/2025 | <p>Today, I implemented an OTP (One-Time Password) system and initiated the Stripe payment gateway integration for Mega Power Gym.</p> <p>In the morning, I developed an SMS-based OTP verification system for secure actions such as payment confirmation and accessing sensitive data. The system generates a secure 6-digit OTP with a 5-minute expiration, includes phone number normalization (+94 format), rate limiting, attempt restrictions, and automatic OTP invalidation. A user-friendly OTP input interface with countdown timer was also implemented. For development, a fallback testing mechanism was added.</p> <p>In the afternoon, I evaluated payment gateway options and selected Stripe due to its strong documentation, international support, and security features. I studied Stripe's payment flow (Payment Intents, 3D Secure, Webhooks) and prepared the database schema updates required for payment tracking. Stripe libraries were installed in both backend and frontend, and detailed payment flow documentation was created to guide implementation.</p> <p><b>Skills Applied:</b><br/>Multi-factor authentication, secure OTP generation, phone number normalization, rate limiting, payment gateway evaluation, Stripe API architecture study, database schema planning.</p>                                                                                                                                          |
| 05/11/2025 | Poya Holiday                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 06/11/2025 | <p>Today, I completed the full payment system integration and optimized the workflow for both card payments and bank transfers.</p> <p><b>Key Tasks Completed:</b></p> <ul style="list-style-type: none"> <li>• Integrated Stripe payments with card processing, 3D Secure support, and automatic membership activation via webhooks.</li> <li>• Developed backend APIs for creating payment intents, confirming transactions, and handling Stripe webhook events with signature verification.</li> <li>• Implemented bank transfer uploads: <ul style="list-style-type: none"> <li>○ Secure file upload using Multer (JPEG, PNG, PDF; max 5MB)</li> <li>○ Drag-and-drop upload interface</li> <li>○ Payments marked as "Pending" until staff approval</li> <li>○ Approval/rejection workflow with detailed status tracking and audit logs</li> </ul> </li> <li>• Performed system testing and optimization: <ul style="list-style-type: none"> <li>○ Fixed prorated package calculation errors</li> <li>○ Resolved concurrent payment approval conflicts with database transactions and row-level locking</li> <li>○ Prevented webhook race conditions</li> <li>○ Improved frontend performance with pagination and React optimizations</li> </ul> </li> <li>• Enhanced security and reliability: <ul style="list-style-type: none"> <li>○ File MIME-type validation</li> <li>○ Controlled email notifications for payment updates</li> </ul> </li> </ul> |
| 07/11/2025 | <p>Today I developed a complete real-time chat system to improve communication between gym members and trainers for workout guidance and support.</p> <p>The system uses hybrid polling (every 3 seconds) and a structured messages table with read status, timestamps, attachments, and soft delete functionality.</p> <p>Frontend features include conversation lists with unread badges, chat bubbles with read receipts, typing indicators, online status, emoji reactions, and message pinning.</p> <p>File sharing supports images, PDFs, and videos (up to 10MB) with drag-and-drop upload, thumbnails, and secure time-limited access links.</p> <p>Additional features include message search, PDF export, infinite scroll, message threading, reporting tools, and automatic archiving of old messages.</p> <p>Performance was optimized with database indexing, and security was ensured through strict authorization and protected file access.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

## SUPERVISOR COMMENTS FOR THE WEEK

*Gamage demonstrated excellent backend and security skills this week. She successfully implemented JWT-based authentication, a secure password reset mechanism using Nodemailer, and an SMS-based OTP system. Furthermore, she efficiently handled the Stripe payment gateway integration—including webhooks and bank transfers—and successfully developed a real-time chat system for member-trainer communication.*

Supervisor's  
Signature



Date

11/14/2025

## Form I – 3A

## INTERN'S DAILY DIARY

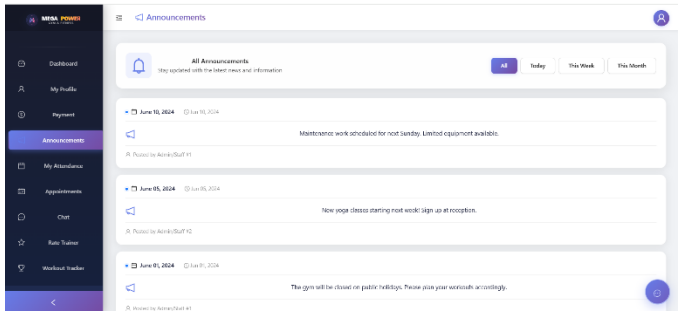
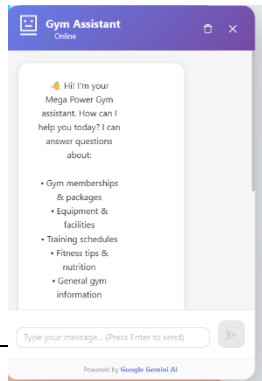
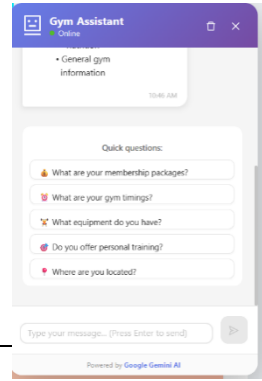
(To be filled by the Intern- Please ensure to upload duly filled set of forms at end of four weeks to the provided folder)

## Intern's Information

Intern's Name **D.N Gamage**Student ID **IT23481244**Internship Title **Software Engineer (Intern)**Specialisation **Information Technology**Supervisor Name **Bandara K.M.V.T**

## Internship Information

## Training Information For the Week (to be filled by the intern)

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|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10/11/2025 | Academic Holiday (Final Exam)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 11/11/2025 | <p>Today I implemented two key communication features: an intelligent chatbot for automated member support and a comprehensive announcement broadcasting system.</p> <p>The chatbot uses a hybrid approach with rule-based intent matching and keyword detection, supported by a database-stored Q&amp;A knowledge base. It includes context-aware responses, transactional queries (e.g., membership status), quick-reply buttons, and smooth handoff to human staff. Analytics track answered, escalated, and common queries for continuous improvement.</p> <p>The announcement system allows staff to create rich-text announcements with priority levels (high/medium/low), target audiences (all members or specific segments), and expiry dates. Features include draft mode, version history, read tracking, automatic email notifications for high-priority posts, scheduled publishing, and auto-expiry.</p> <p>Security, performance, and usability were enhanced through content sanitization (XSS prevention), targeted filtering logic, background jobs, timezone handling, and analytics dashboards.</p> <div>    </div> |

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|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12/11/2025 | <p>Today I focused on developing and enhancing the Member Portal, Profile Management, and Registration System, ensuring a seamless and secure experience for gym members from registration to daily usage.</p> <p>I designed the Member Dashboard (MemberDashboard.jsx) as a fully personalized interface where members can conveniently view their membership status, upcoming appointments, attendance history, announcements, payment records, and assigned workout plans. The dashboard centralizes all essential information, improving user engagement and accessibility.</p> <p>For Member Profile Management (MemberProfile.jsx), I implemented comprehensive profile editing features, including profile photo upload and secure handling of sensitive updates. Re-authentication is required for critical changes, along with email verification for added account security. Activity logs were introduced to track profile-related actions, and a GDPR-compliant data export feature enables members to download their personal data when required.</p> <p>The Enhanced Registration System (MemberRegister.jsx) was developed as a structured multi-step wizard to simplify onboarding. It includes membership package selection with dynamic pricing calculations, integrated payment options (Stripe and bank transfer), duplicate account prevention, and Google Places API integration for accurate address input. Upon successful registration, the system automatically sends a welcome email containing a unique QR code for identification and attendance tracking.</p> <p>Finally, I conducted comprehensive end-to-end testing covering registration workflows, form validations, payment integrations, and system edge cases. Several minor bugs were identified and resolved to ensure stability, security, and smooth user experience across all modules.</p> |
| 13/11/2025 | Academic Holiday (Mid-Exam)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 14/11/2025 | Academic Holiday (Final Exam)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

## SUPERVISOR COMMENTS FOR THE WEEK

*Despite a shortened week due to academic exams, Gamage delivered highly impactful features. She successfully built an intelligent chatbot for automated member support and an advanced announcement broadcasting system with targeted segments. She also finalized the Member Portal and enhanced the registration workflow, seamlessly integrating the Google Places API and automated QR-code welcome emails.*

Supervisor's  
Signature



Date

11/21/2025

# Form I – 3A

## INTERN'S DAILY DIARY

*(To be filled by the Intern- Please ensure to upload duly filled set of forms at end of four weeks to the provided folder)*

### Intern's Information

Intern's Name **D.N Gamage**

Student ID **IT23481244**

Internship Title **Software Engineer (Intern)**

Specialisation **Information Technology**

Supervisor  
Name **Bandara K.M.V.T**

### Internship Information

#### Training Information For the Week (to be filled by the intern)

| DATE       | DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., SKETCHES AND DIMENSIONS TO BE GIVEN WHEREVER POSSIBLE. |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 17/11/2025 | Academic Leave (Final Exam preparation)                                                                                                 |
| 18/11/2025 | Academic Holiday (Final Exam)                                                                                                           |

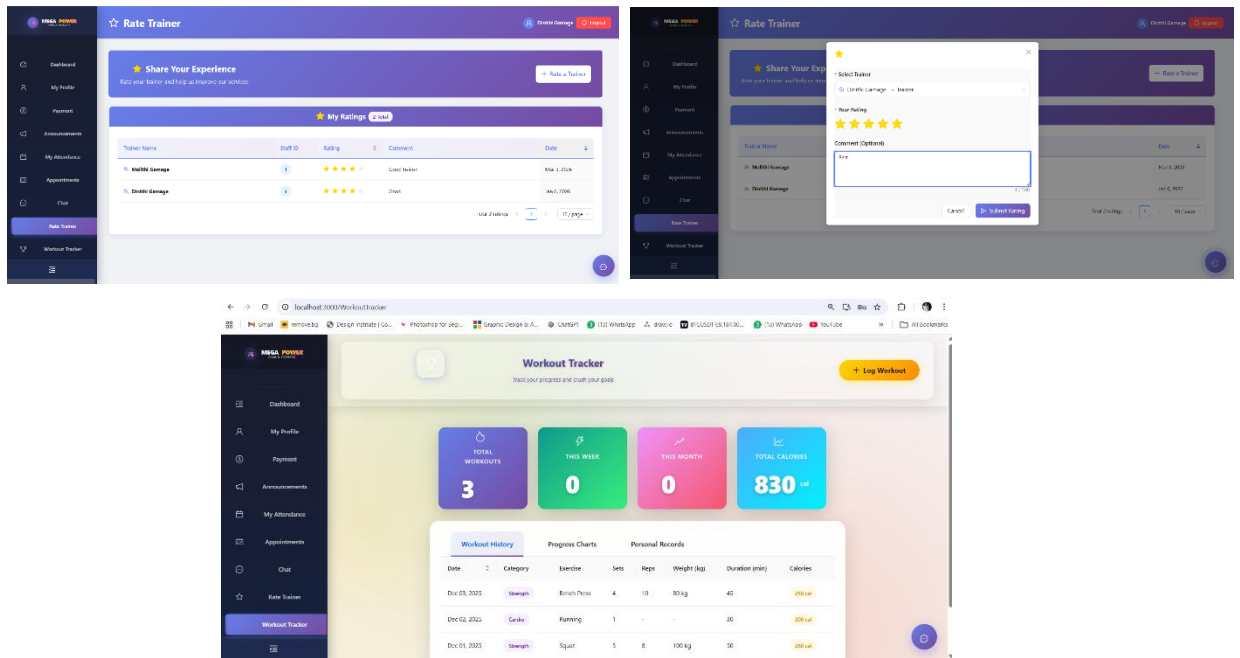
19/11/2025

Today I implemented a comprehensive rating and feedback system for trainers and classes, alongside enhanced workout plan tracking for members.

The rating system allowed members to submit 1–5 star ratings with optional reviews via `MemberFeedback.jsx`, validated by attendance history and cooldown periods to prevent spam. Ratings could be anonymous, edited within 24 hours, and aggregated in `Feedbacktable.jsx` for admins and trainers. Low ratings triggered management alerts, while analytics provided insights on trends, top-rated trainers, and member satisfaction. Public pages displayed average ratings for trainers and classes. Workout plan tracking allowed trainers to create structured programs stored in `WorkoutPlan` and `WorkoutExercises` tables. Members viewed plans in `MemberDashboard.jsx`, marking exercises completed and logging actual performance. Progress charts visualized adherence and strength gains, with automatic reminders for missed workouts. Trainers could monitor member progress, modify plans with versioning, and integrate achievements and badges for motivation.

#### Key Skills:

Database constraints and aggregation, React star-rating components, form validation, notifications, `Chart.js` visualizations, versioning, cron-based reminders, and integrated data modeling for workouts and attendance.



20/11/2025

Today I redesigned the Mega Power Gym interface with a modern dark theme. I updated the login page, made a collapsible sidebar, and improved the admin dashboard with interactive charts. The new theme works across 30+ components with good contrast, responsive layouts, and smooth animations. I also improved accessibility with keyboard navigation, screen reader support, and clear focus indicators. Performance was made faster using code-splitting, lazy loading, and optimized images.

Today, Mr. K.M.V.T. Bandara also conducted a meeting to review the progress.



|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 21/11/2025 | <p>Today marked the completion of the Mega Power Gym project. Mr. K.M.V.T. Bandara conducted a meeting to review all functionalities and ensure everything worked correctly. The day involved final enhancements, code refactoring, performance optimization, testing, and deployment preparation.</p> <p>I focused on building advanced reusable UI components including Buttons (variants, sizes, loading states), Cards (header, body, footer, hover/collapse), Modals (focus trap, animations, scroll lock), DataTables (sorting, pagination, search, CSV export), Inputs, Selects, DatePickers, Tooltips, and Toast notifications. These components-maintained consistency across 30+ views and improved development efficiency.</p> <p>Code and CSS were refactored for maintainability and performance using Context API, custom hooks, memoization, BEM, and theming. Backend and database optimizations improved security, validation, and query efficiency.</p> <p>Comprehensive testing (unit, integration, E2E, manual) fixed critical bugs, and performance enhancements (code-splitting, lazy loading, caching) achieved high Lighthouse scores. Deployment scripts and staging validation ensured the app was production ready.</p> |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## SUPERVISOR COMMENTS FOR THE WEEK

*Gamage successfully concluded the Mega Power Gym project this week. She implemented a comprehensive rating system and an advanced workout plan tracker with Chart.js visualizations. I am particularly impressed with her final UI overhaul, which introduced a responsive dark theme, and her dedication to code refactoring, performance optimization, and rigorous deployment preparation.*

Supervisor's  
Signature



Date

12/5/2025

## Form I – 3A

## INTERN'S DAILY DIARY

(To be filled by the Intern- Please ensure to upload duly filled set of forms at end of four weeks to the provided folder)

## Intern's Information

Intern's Name

D.N Gamage

Student ID

IT23481244

Internship Title

Software Engineer (Intern)

Specialisation

Information Technology

Supervisor  
Name

Bandara K.M.V.T

## Internship Information

## Training Information For the Week (to be filled by the intern)

| DATE       | DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., SKETCHES AND DIMENSIONS TO BE GIVEN WHEREVER POSSIBLE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 24/11/2025 | <p>My supervisor assigns me to join the software development team for the POS web application project. My team lead, H.M.L.P.Abayaratne, provided an overview of the project, its objectives, and expected deliverables. I spent the day understanding my assigned tasks, familiarizing myself with the project structure, and coordinating with other team members.</p> <p>Goals for the day:</p> <ul style="list-style-type: none"> <li>• Understand the POS project scope</li> <li>• Learn my responsibilities as a team member</li> <li>• Get familiar with the team and their roles</li> </ul>                                                                                                                                                                                                                                                                                                   |
| 25/11/2025 | <p>Today, I participated in the first internal team meeting for the POS project. The team discussed roles and initial task assignments:</p> <ul style="list-style-type: none"> <li>• Frontend development: UI pages for login, dashboard, and billing</li> <li>• Backend development: Database setup and API endpoints</li> <li>• UI/UX design: Wireframes and mobile responsiveness</li> </ul> <p>I also spent time reviewing existing POS systems to understand features and best practices we could implement in our project.</p> <p>Some challenges included understanding the client's expectations without direct communication and ensuring everyone on the team had the same understanding of the project goals.</p> <p>Key Learning:</p> <p>Early coordination and clear task distribution are important, and researching similar systems beforehand helps in future client discussions.</p> |

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 26/11/2025 | <p>Today, I joined my supervisor in a meeting with the client to clarify the functional requirements for the new POS web application.</p> <p>The discussion covered:</p> <ul style="list-style-type: none"> <li>• Core functions: Billing, inventory management, and reporting</li> <li>• User roles: Cashier, Manager, and Administrator</li> <li>• Key priorities: Mobile responsiveness and ease of use</li> </ul> <p>The client shared specific challenges with their current system, which gave us clear direction for our development tasks.</p> <p>Key Learning: Direct client input is essential for understanding requirements accurately and helps reduce revisions later. The insights from this meeting will guide our team's development approach and planning.</p>                                                                                                                                                     |
| 27/11/2025 | <p>Following yesterday's client meeting, I participated in an internal analysis session with the development team to translate the gathered requirements into actionable plans.</p> <p>Progress Made:</p> <ul style="list-style-type: none"> <li>• Helped refine and document the core functional modules: Billing, Inventory, Product Management, Reports, and User Roles.</li> <li>• Assisted in identifying key technical requirements such as barcode scanning functionality and configurable discount/pricing rules.</li> <li>• Contributed to drafting the initial User Stories and Acceptance Criteria for the prioritized features.</li> </ul> <p>Key Learning: Breaking down high-level client needs into structured, detailed requirements is essential. Clear requirements enable the team to work efficiently and ensure our output aligns with the client's expectations.</p>                                           |
| 28/11/2025 | <p>Following the detailed requirement analysis, I worked with the development team to formalize the requirements into structured documentation.</p> <p>Progress Made:</p> <ul style="list-style-type: none"> <li>• Assisted in consolidating the refined requirements into the company's System Requirements Specification (SRS) format.</li> <li>• Helped draft high-level requirement descriptions and expanded the initial list of user stories for the core modules.</li> <li>• Contributed to compiling details for the technical architecture diagrams and the final project proposal.</li> <li>• Participated in a progress check-in with the team to ensure alignment and address any immediate blockers.</li> </ul> <p>Key Learning: Good documentation serves as the team's shared blueprint. Investing time in clear documentation now prevents confusion and rework later, ensuring everyone builds the same vision.</p> |

## SUPERVISOR COMMENTS FOR THE WEEK

*Transitioning to the new POS web application project, Gamage showed strong adaptability and communication skills. She actively participated in internal team discussions and client meetings to effectively gather core functional requirements. Additionally, she played a valuable role in translating these business needs into technical documentation, User Stories, and the formal System Requirements Specification (SRS).*

Supervisor's  
Signature



Date

12/5/2025



## Form I – 3A

## INTERN'S DAILY DIARY

(To be filled by the Intern- Please ensure to upload duly filled set of forms at end of four weeks to the provided folder)

## Intern's Information

Intern's Name

D.N Gamage

Student ID

IT23481244

Internship Title

Software Engineer (Intern)

Specialisation

Information Technology

Supervisor  
Name

Bandara K.M.V.T

## Internship Information

## Training Information For the Week (to be filled by the intern)

| DATE       | DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., SKETCHES AND DIMENSIONS TO BE GIVEN WHEREVER POSSIBLE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01/12/2025 | <p>Today, I was involved in both project coordination and backend development tasks. It was a valuable experience balancing planning responsibilities with technical work.</p> <p>Progress Made:</p> <p>Project Coordination:</p> <ul style="list-style-type: none"> <li>Participated in defining the roadmap for the upcoming module.</li> <li>Helped establish clear project phases.</li> <li>Clarified two main operational workflows:               <ul style="list-style-type: none"> <li><b>Retail Mode</b> – Standard point-of-sale process.</li> <li><b>Restaurant Mode</b> – Includes table management and order handling.</li> </ul> </li> <li>These workflows will shape the overall system logic.</li> </ul> <p>Backend Development:</p> <ul style="list-style-type: none"> <li>Contributed ideas for system scalability and architecture planning.</li> <li>Worked with another backend developer to draft the initial API endpoint structure.</li> <li>Reviewed the core database schema to ensure it supports the defined business workflows.</li> </ul> |

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 02/12/2025 | <p>Today, I participated in a detailed discussion to gather functional requirements for the restaurant-specific workflow of the POS system. After the meeting, I helped create a visual workflow diagram using Draw.io to represent the Restaurant POS processes clearly.</p> <p>Progress Made:</p> <ul style="list-style-type: none"> <li>Contributed to developing the Restaurant POS workflow diagram, including: <ul style="list-style-type: none"> <li>Table assignment and management</li> <li>Order routing to the kitchen and bar</li> <li>Kitchen ticket generation and status updates</li> <li>Separate workflows for dine-in and take-away orders</li> </ul> </li> <li>Supported the UI/UX team by providing a clear workflow structure to help them design restaurant-specific screens.</li> </ul> |
| 03/12/2025 | <p>Today, I took part in a detailed discussion to gather functional requirements for the restaurant-specific workflow of the POS system. After the session, I created a visual diagram using Draw.io to clearly model the connected processes of the Restaurant POS system.</p> <p>Progress Made:</p> <ul style="list-style-type: none"> <li>Developed a Restaurant POS workflow diagram including: <ul style="list-style-type: none"> <li>Table assignment and management</li> <li>Order routing to the kitchen and bar</li> <li>Kitchen ticket generation and status updates</li> <li>Separate workflows for dine-in and take-away orders</li> </ul> </li> <li>Supported the UI/UX team by providing a clear workflow structure to design user-friendly restaurant screens.</li> </ul>                       |
| 04/12/2025 | Poya Holiday                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 05/12/2025 | <p>Today, I focused on understanding the overall Restaurant POS system and preparing the backend environment.</p> <ul style="list-style-type: none"> <li>Studied system requirements and workflows, covering: <ul style="list-style-type: none"> <li>Sales processing</li> <li>Inventory management</li> <li>Kitchen operations</li> <li>User roles and permissions</li> </ul> </li> <li>Set up the backend development environment with Node.js and Express.</li> <li>Verified that the initial server setup and API endpoints were running correctly.</li> </ul>                                                                                                                                                                                                                                             |

## SUPERVISOR COMMENTS FOR THE WEEK

*Gamage had a highly productive start to the month, effectively balancing project coordination with backend preparations. She played a crucial role in gathering functional requirements and successfully translated them into a clear Restaurant POS workflow diagram using Draw.io to assist the UI/UX team. Her proactive setup of the Node.js and Express backend environment demonstrates excellent initiative and readiness for the development phase.*

Supervisor's  
Signature



Date 12/12/2025

## Form I – 3A

## INTERN'S DAILY DIARY

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## Intern's Information

Intern's Name

D.N Gamage

Student ID

IT23481244

Internship Title

Software Engineer (Intern)

Specialisation

Information Technology

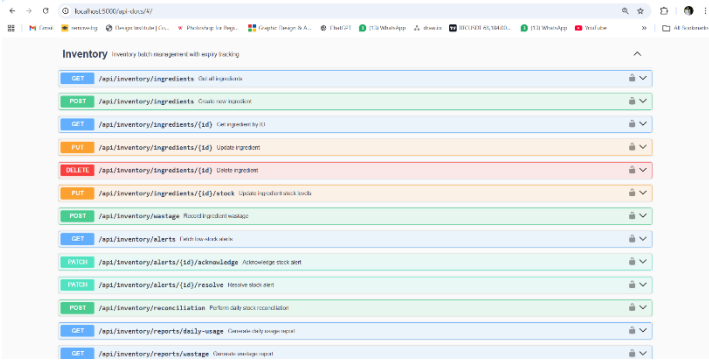
Supervisor  
Name

Bandara K.M.V.T

## Internship Information

## Training Information For the Week (to be filled by the intern)

| DATE       | DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., SKETCHES AND DIMENSIONS TO BE GIVEN WHEREVER POSSIBLE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 08/12/2025 | <p>Today, I started hands-on database design work with guidance from the senior backend developer to finalize the core schema for the POS system.</p> <p>Progress Made:</p> <ul style="list-style-type: none"> <li>Designed the initial ER diagram covering:               <ul style="list-style-type: none"> <li>Users and authentication</li> <li>Products and inventory</li> <li>Sales transactions</li> <li>Restaurant tables and orders</li> <li>Payment records</li> </ul> </li> <li>Defined key relationships (one-to-many and many-to-many).</li> <li>Created migration scripts for users, products, and categories tables.</li> <li>Configured the MySQL database connection in the development environment.</li> </ul> <p>Technical Stack:</p> <ul style="list-style-type: none"> <li>Database: MySQL 8.0</li> </ul> |
| 09/12/2025 | <p>Today, I focused on setting up the backend structure and implementing the authentication system.</p> <p>Progress Made:</p> <ul style="list-style-type: none"> <li>Initialized a Node.js project with Express.</li> <li>Organized the project into folders: controllers, models, routes, middleware, and config.</li> <li>Implemented JWT-based authentication:               <ul style="list-style-type: none"> <li>User registration and login endpoints</li> <li>Password hashing using bcrypt</li> <li>Token generation and validation middleware</li> </ul> </li> <li>Created the User model using Sequelize ORM.</li> <li>Tested APIs using Postman.</li> </ul>                                                                                                                                                        |

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10/12/2025 | <p>Today, I worked on implementing Role-Based Access Control (RBAC) and developing product management endpoints. I also attended a meeting with my supervisor and team lead to review progress and clarify next steps.</p> <p>Progress Made:</p> <ul style="list-style-type: none"> <li>Updated the User model to include roles (Admin, Manager, Cashier, Waiter).</li> <li>Created authorization middleware to restrict access based on user roles.</li> <li>Developed Product CRUD APIs: <ul style="list-style-type: none"> <li>Create product (Admin/Manager only)</li> <li>View products (All authorized users)</li> <li>Update product (Admin/Manager only)</li> <li>Delete product (Admin only)</li> </ul> </li> <li>Created Product and Category models with proper associations.</li> <li>Tested endpoints with different roles to ensure correct access control.</li> </ul>                                                                                                                                                                                     |
| 11/12/2025 | <p>Today, I worked on developing the inventory management system and stock tracking features.</p> <p>Progress Made:</p> <ul style="list-style-type: none"> <li>Created the Inventory model with: <ul style="list-style-type: none"> <li>Product reference</li> <li>Current stock quantity</li> <li>Minimum and maximum stock levels</li> <li>Last updated timestamp</li> </ul> </li> <li>Developed inventory APIs: <ul style="list-style-type: none"> <li>Stock-in (add stock)</li> <li>Stock-out (reduce stock)</li> <li>Low-stock product listing</li> <li>Inventory report generation</li> </ul> </li> <li>Added automatic stock validation during sales creation.</li> <li>Implemented a low-stock alert system.</li> <li>Set up database triggers for automatic stock updates.</li> </ul>                                                                                                                                                                                       |
| 12/12/2025 | <p>Today, I worked on developing the sales transaction system, which is the main feature of the POS system. I also attended a meeting with Ms. ABAYARATHNE H. M. L. P and we discussed the development approach and worked together on key parts of the implementation.</p> <ul style="list-style-type: none"> <li>Created a Sales model including: <ul style="list-style-type: none"> <li>Auto-generated transaction ID</li> <li>Cashier/waiter reference</li> <li>Customer details</li> <li>Sale items (products with quantities)</li> <li>Subtotal, tax (15% VAT for Sri Lanka), discount, and total amount</li> <li>Payment method and status</li> <li>Timestamp</li> </ul> </li> <li>Implemented sales APIs: <ul style="list-style-type: none"> <li>Create new sale</li> <li>View all sales with filters</li> <li>View sale details</li> <li>Void a sale</li> </ul> </li> <li>Added automatic inventory deduction when a sale is created.</li> <li>Created a SaleItem junction model to manage the many-to-many relationship between sales and products.</li> </ul> |



- Used Sequelize transactions to ensure all steps (sale creation, sale items, inventory update) are completed successfully or rolled back if an error occurs.



## SUPERVISOR COMMENTS FOR THE WEEK

*This week, Gamage demonstrated strong technical capabilities in database and API design. She successfully finalized the core schema ER diagram for MySQL 8.0 and implemented a secure JWT-based authentication system with password hashing. Furthermore, she effectively developed Role-Based Access Control (RBAC), an inventory management module with automatic stock validation, and the core sales transaction system utilizing Sequelize transactions. Her collaborative efforts with Ms. Abayarathne on the sales implementation were highly commendable.*

Supervisor's  
Signature



Date

12/19/2025

## Form I – 3A

## INTERN'S DAILY DIARY

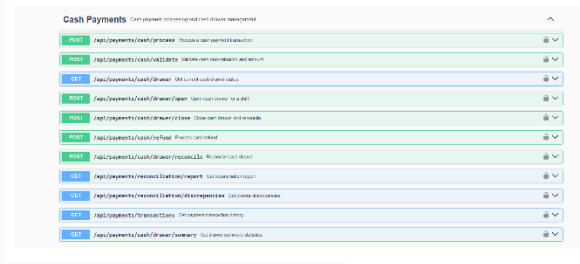
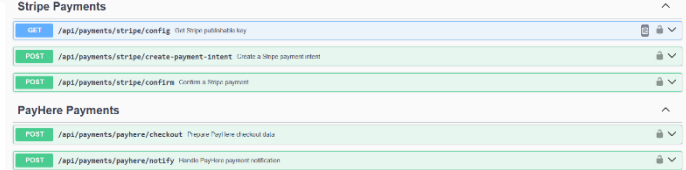
(To be filled by the Intern- Please ensure to upload duly filled set of forms at end of four weeks to the provided folder)

## Intern's Information

Intern's Name **D.N Gamage**Student ID **IT23481244**Internship Title **Software Engineer (Intern)**Specialisation **Information Technology**Supervisor Name **Bandara K.M.V.T**

## Internship Information

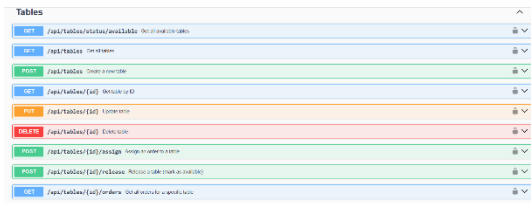
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|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15/12/2025 | <p>Today, I worked on adding multiple payment options and implementing receipt generation for the POS system.</p> <ul style="list-style-type: none"> <li>Updated the Payment model to support:             <ul style="list-style-type: none"> <li>Cash payments</li> <li>Card payments (Stripe integration)</li> <li>Mobile payments (PayHere – Sri Lanka)</li> <li>Split payments (using more than one payment method for a single sale)</li> </ul> </li> <li>Developed payment APIs for processing cash, card, mobile, and split payments.</li> <li>Created a receipt generation service:             <ul style="list-style-type: none"> <li>Generates PDF receipts</li> <li>Includes a QR code for digital access</li> <li>Added option to send receipts via email</li> </ul> </li> <li>Integrated Stripe test API for card transactions.</li> </ul> <div>    </div> |

16/12/2025

Today, I worked on developing the restaurant table management feature for the POS system.  
Progress Made:

- Created a Table model with:
  - Table number
  - Seating capacity
  - Current status (Available, Occupied, Reserved, Cleaning)
  - Assigned waiter
  - Location/section
- Developed table management APIs:
  - Create table
  - View all tables with current status
  - Update table status
  - Assign waiter to a table
- Designed a data structure for table layout visualization.
- Implemented a table reservation system with time slot management.

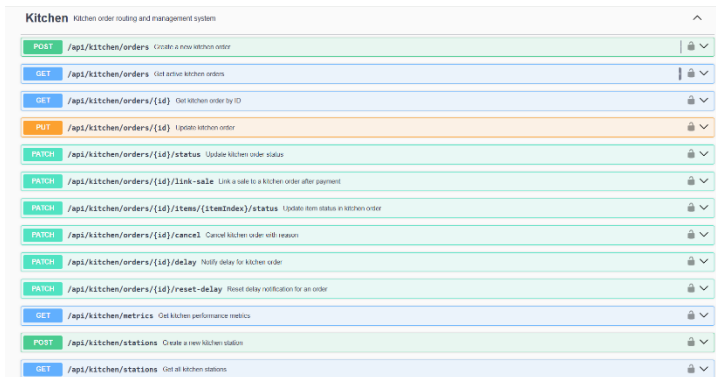


| Method | Endpoint                     | Description                         |
|--------|------------------------------|-------------------------------------|
| GET    | /api/tables/status/available | Get available tables                |
| GET    | /api/tables                  | Get tables                          |
| POST   | /api/tables                  | Create a new table                  |
| GET    | /api/tables/{id}             | Get table by ID                     |
| PUT    | /api/tables/{id}             | Update table                        |
| DELETE | /api/tables/{id}             | Delete table                        |
| POST   | /api/tables/{id}/assign      | Assign a waiter to a table          |
| POST   | /api/tables/{id}/reserve     | Reserve a table for a specific time |
| GET    | /api/tables/{id}/orders      | Get all orders for a specific table |

17/12/2025

Today, I supported Ms. H.M.L.P. Abayarathne in developing the kitchen order management and ticket system for restaurant operations. We collaborated to implement this module together

- Created KitchenOrder model with:
  - Order number
  - Table reference
  - Items ordered
  - Priority level
  - Preparation status (Pending, Preparing, Ready, Served)
  - Timestamps for each status change
- Developed kitchen endpoints:
  - Create kitchen orders
  - View pending orders
  - Update order status
  - View ready/completed orders
- Added automatic order routing:
  - Food → Kitchen
  - Beverages → Bar
- Designed a Kitchen Display System (KDS) data structure.
- Implemented order priority system to manage preparation times.



| Method | Endpoint                                          | Description                                  |
|--------|---------------------------------------------------|----------------------------------------------|
| POST   | /api/kitchen/orders                               | Create a new kitchen order                   |
| GET    | /api/kitchen/orders                               | Get kitchen orders                           |
| GET    | /api/kitchen/orders/{id}                          | Get kitchen order by ID                      |
| PUT    | /api/kitchen/orders/{id}                          | Update kitchen order                         |
| PATCH  | /api/kitchen/orders/{id}/status                   | Update kitchen order status                  |
| PATCH  | /api/kitchen/orders/{id}/link-sale                | Link a sale to a kitchen order after payment |
| PATCH  | /api/kitchen/orders/{id}/items/{itemIndex}/status | Update item status in kitchen order          |
| PATCH  | /api/kitchen/orders/{id}/cancel                   | Cancel kitchen order with reason             |
| PATCH  | /api/kitchen/orders/{id}/delay                    | Notify delay for kitchen order               |
| PATCH  | /api/kitchen/orders/{id}/reset-delay              | Reset delay notification for an order        |
| GET    | /api/kitchen/metrics                              | Get kitchen performance metrics              |
| POST   | /api/kitchen/stations                             | Create a new kitchen station                 |
| GET    | /api/kitchen/stations                             | Get all kitchen stations                     |

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 18/12/2025 | <p>Today, I worked on integrating the full restaurant workflow from order to payment, ensuring smooth coordination between tables, kitchen, and billing.</p> <p>Progress Made:</p> <ul style="list-style-type: none"> <li>• Connected table management with order creation.</li> <li>• Implemented dine-in workflow: <ol style="list-style-type: none"> <li>1. Customer seated → Table marked as Occupied</li> <li>2. Waiter takes order → Kitchen order created</li> <li>3. Kitchen prepares → Status visible to waiter</li> <li>4. Food served → Items marked as served</li> <li>5. Bill requested → Sale generated from order items</li> <li>6. Payment processed → Table marked Available</li> </ol> </li> <li>• Developed order modification system: add/remove items, change quantities, with kitchen notifications.</li> <li>• Added order splitting for shared bills.</li> <li>• Tracked customer count per table.</li> </ul> |
| 19/12/2025 | <p>Today, I focused on building the reporting and analytics system for the POS. My supervisor, Mr. K.M.V.T. Bandara, also conducted a meeting with the development team to review progress.</p> <ul style="list-style-type: none"> <li>• Implemented sales reports: <ul style="list-style-type: none"> <li>○ Daily sales summary</li> <li>○ Product performance</li> <li>○ Staff performance</li> <li>○ Payment method breakdown</li> <li>○ Peak hours analysis</li> </ul> </li> <li>• Created inventory reports: <ul style="list-style-type: none"> <li>○ Current stock levels</li> <li>○ Fast- and slow-moving items</li> <li>○ Stock value calculation</li> </ul> </li> <li>• Added date range filtering for all reports.</li> <li>• Aggregated data using Sequelize queries.</li> <li>• Enabled export in CSV, Excel, and PDF formats.</li> </ul>                                                                                 |

## SUPERVISOR COMMENTS FOR THE WEEK

*Gamage made significant strides in feature implementation this week. She successfully integrated diverse payment options, including Stripe and PayHere, alongside a PDF receipt generation service. She also developed a comprehensive table management system and a kitchen order routing system complete with priority levels and automated routing based on item type. Her ability to seamlessly connect the entire end-to-end dine-in workflow—from seating to billing—is impressive. Additionally, she built a robust reporting and analytics module capable of exporting data in CSV, Excel, and PDF formats.*

Supervisor's  
Signature



Date

12/26/2025

## Form I – 3A

## INTERN'S DAILY DIARY

(To be filled by the Intern- Please ensure to upload duly filled set of forms at end of four weeks to the provided folder)

## Intern's Information

Intern's Name

D.N Gamage

Student ID

IT23481244

Internship Title

Software Engineer (Intern)

Specialisation

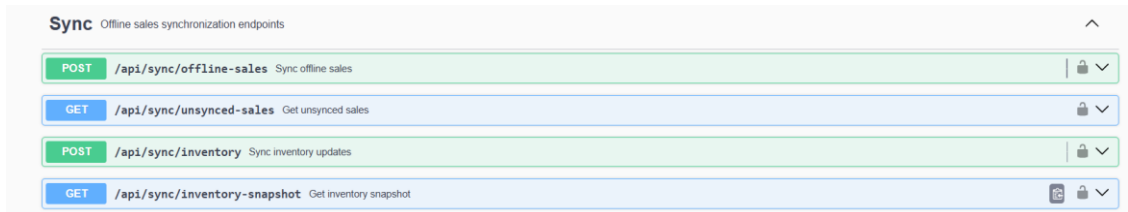
Information Technology

Supervisor  
Name

Bandara K.M.V.T

## Internship Information

## Training Information For the Week (to be filled by the intern)

| DATE       | DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., SKETCHES AND DIMENSIONS TO BE GIVEN WHEREVER POSSIBLE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 22/12/2025 | <p>Today, I implemented offline-first support for the POS system.</p> <p>Progress Made:</p> <ul style="list-style-type: none"> <li>Designed offline data sync architecture.</li> <li>Used IndexedDB for local storage of transactions and data.</li> <li>Implemented sync endpoints:               <ul style="list-style-type: none"> <li>Pull latest data from server (POST /api/sync/pull)</li> <li>Push local changes to server (POST /api/sync/push)</li> <li>Check sync status (GET /api/sync/status)</li> </ul> </li> <li>Added conflict resolution for offline edits.</li> <li>Created transaction queue for offline operations.</li> <li>UI shows sync status and automatically syncs when online.</li> </ul> <p><b>Technical Implementation:</b></p> <ul style="list-style-type: none"> <li>Timestamp-based conflict resolution.</li> <li>Optimistic locking for concurrent updates.</li> <li>Sync logs maintained for auditing.</li> </ul>  <p>The screenshot shows a list of API endpoints under the heading 'Sync Offline sales synchronization endpoints'. There are four entries: a POST endpoint for syncing offline sales, a GET endpoint for getting unsynced sales, a POST endpoint for syncing inventory updates, and a GET endpoint for getting an inventory snapshot. Each entry includes the HTTP method, the endpoint path, a brief description, and a lock icon.</p> |

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 23/12/2025 | <p>Today, I focused on improving API documentation and making it easier for developers to use our POS APIs.</p> <p>Progress Made:</p> <ul style="list-style-type: none"> <li>• Integrated Swagger/OpenAPI for interactive API docs.</li> <li>• Documented all endpoints with: <ul style="list-style-type: none"> <li>○ Request parameters and body schemas</li> <li>○ Response formats and status codes</li> <li>○ Authentication details</li> <li>○ Example requests/responses</li> </ul> </li> <li>• Hosted interactive docs at /api-docs.</li> <li>• Added detailed error response formats.</li> <li>• Prepared API usage guide for frontend developers.</li> <li>• Created a Postman collection covering all endpoints.</li> </ul>                                      |
| 24/12/2025 | Holiday                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 25/12/2025 | Holiday                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 26/12/2025 | <p>Today, I worked closely with the frontend team to integrate the first set of backend APIs with the POS frontend.</p> <ul style="list-style-type: none"> <li>• Held integration meeting with frontend developers.</li> <li>• Resolved CORS issues and standardized API responses.</li> <li>• Implemented request/response logging for debugging.</li> <li>• Added consistent error handling via middleware.</li> <li>• Validated requests using express-validator.</li> <li>• Fixed bugs found during integration: <ul style="list-style-type: none"> <li>○ Added pagination to list endpoints</li> <li>○ Standardized date formats in responses</li> <li>○ Ensured required fields are returned</li> </ul> </li> <li>• Tested authentication flow end-to-end.</li> </ul> |

## SUPERVISOR COMMENTS FOR THE WEEK

*Gamage showed excellent technical initiative this week by implementing offline-first support for the POS system, utilizing IndexedDB for local storage and developing robust sync endpoints with conflict resolution. She also significantly improved the project's developer experience by integrating Swagger/OpenAPI for interactive API documentation and creating a comprehensive Postman collection. Furthermore, her close collaboration with the frontend team to integrate APIs, resolve CORS issues, and standardize error handling ensured a smooth and secure connection between the systems.*

Supervisor's  
Signature



Date 1/9/2026

## Form I – 3A

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## Intern's Information

Intern's Name

D.N Gamage

Student ID

IT23481244

Internship Title

Software Engineer (Intern)

Specialisation

Information Technology

Supervisor  
Name

Bandara K.M.V.T

## Internship Information

## Training Information For the Week (to be filled by the intern)

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|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 29/12/2025 | <p>Today, I enhanced the POS system with advanced search and filtering for products.</p> <ul style="list-style-type: none"> <li>Added search by name, category, SKU, and barcode.</li> <li>Enabled filters: price range, stock availability.</li> <li>Added sorting options: price, name, popularity.</li> <li>Implemented pagination for all list endpoints.</li> <li>Used MySQL FULLTEXT indexes for fast full-text search.</li> <li>Created autocomplete suggestions endpoint.</li> <li>Optimized queries with proper indexing and monitored performance.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 30/12/2025 | <p>Today, I worked on adding bill splitting functionality for restaurant operations.</p> <p>Progress Made:</p> <ul style="list-style-type: none"> <li>Created BillSplit model to manage split payments.</li> <li>Implemented multiple splitting strategies:               <ul style="list-style-type: none"> <li>Equal split – Divide bill evenly among N people</li> <li>Item-based split – Each person pays for their ordered items</li> <li>Custom split – Manual allocation of amounts</li> <li>Percentage split – Split based on assigned percentages</li> </ul> </li> <li>Developed endpoints:               <ul style="list-style-type: none"> <li>POST /api/bills/:id/split – Create a split configuration</li> <li>GET /api/bills/:id/splits – View split details</li> <li>POST /api/bills/:id/splits/:splitId/pay – Pay individual portion</li> </ul> </li> <li>Ensured validation so all splits add up to the total bill.</li> <li>Generated individual receipts for each split payment.</li> </ul> |

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31/12/2025 | This day was taken as approved leave due to personal reason.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 01/01/2026 | This day was taken as approved leave due to personal reasons.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 02/01/2026 | <p>Today, the team conducted a full review and testing of the POS system:</p> <ul style="list-style-type: none"> <li>• Performed comprehensive testing of all features: retail and restaurant sales, payments, inventory, and reports.</li> <li>• Documented 15 minor bugs; fixed 8 critical issues immediately, including: <ul style="list-style-type: none"> <li>○ Prevented negative stock in inventory</li> <li>○ Resolved concurrent payment conflicts</li> <li>○ Fixed table status race conditions</li> <li>○ Corrected receipt formatting errors</li> </ul> </li> <li>• Updated task board with remaining fixes.</li> <li>• Compiled year-end progress report: <ul style="list-style-type: none"> <li>○ 45+ API endpoints implemented</li> <li>○ 12 database tables designed</li> <li>○ Authentication &amp; authorization complete</li> <li>○ Core POS functionality fully operational</li> </ul> </li> </ul> |

## SUPERVISOR COMMENTS FOR THE WEEK

*In her final week of the month, Gamage successfully implemented advanced search capabilities using MySQL FULLTEXT indexes and developed complex bill-splitting logic to handle various payment strategies for restaurant operations. Despite taking two days of approved leave, she actively participated in the team's comprehensive end-to-end testing, immediately resolving critical issues such as negative stock prevention and table status race conditions. Her efforts culminated in a highly successful year-end progress report, reflecting a robust and fully operational core POS system.*

Supervisor's  
Signature



Date

1/9/2026



## Form I – 3A

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Supervisor  
Name

Bandara K.M.V.T

## Internship Information

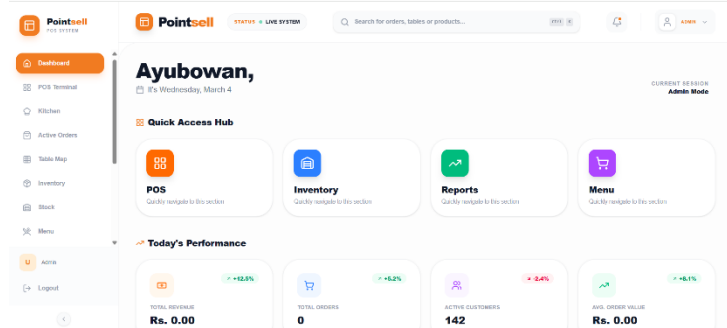
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|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 05/01/2026 | <p>Today, I focused on improving the POS system's security.</p> <p>Progress Made:</p> <ul style="list-style-type: none"> <li>Added rate limiting on login endpoints to prevent brute-force attacks.</li> <li>Implemented Helmet.js for secure HTTP headers.</li> <li>Sanitized inputs to prevent SQL injection and XSS.</li> <li>Created audit logs for critical actions (logins, permission changes, price updates, inventory adjustments, voided transactions).</li> <li>Added encryption for sensitive data.</li> <li>Enforced strong password policies: minimum length, complexity, history, and expiry.</li> <li>Developed security audit endpoints for admins.</li> </ul>                                                                                             |
| 06/01/2026 | <p>Today, I worked on building frontend components in collaboration with the UI/UX team.</p> <p>Progress Made:</p> <ul style="list-style-type: none"> <li>Set up React project with Vite.</li> <li>Built authentication components: login page with validation, role-based dashboards, and protected routes.</li> <li>Developed reusable UI components: buttons, inputs, modals, alerts/notifications.</li> <li>Implemented state management with React Context:               <ul style="list-style-type: none"> <li>AuthContext for user login</li> <li>CartContext for sale items</li> <li>ThemeContext for dark/light mode</li> </ul> </li> <li>Integrated API calls using Axios.</li> <li>Added error boundaries for handling unexpected errors gracefully.</li> </ul> |

07/01/2026

Today, I focused on building the main cashier interface for the POS system.

- Designed cashier dashboard with:
  - Product search and selection
  - Shopping cart display with quantity adjustments
  - Discount and tax calculation
  - Total amount summary
- Integrated barcode scanner for quick item entry.
- Added keyboard shortcuts for common actions.
- Created quick-access grid for popular products.
- Implemented customer info capture.
- Built payment selection and tender/change calculation.
- Managed complex cart state using useReducer hook for efficiency.

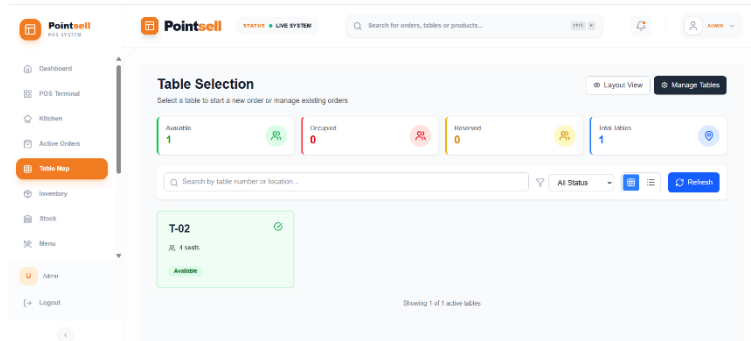


08/01/2026

Today, I worked on building the restaurant table management UI, while Ms. Abayarathne conducted a meeting to evaluate POS system progress.

Progress Made:

- Designed visual table layout:
  - Drag-and-drop arrangement
  - Color-coded status indicators
  - Real-time updates
  - Table capacity display
- Implemented table operations:
  - Status toggle (Available, Occupied, Reserved, Cleaning)
  - Waiter assignment
  - Customer count input and notes
- Added table filter/search and floor/section view for multi-level restaurants.
- Created reservation time-slot viewer.

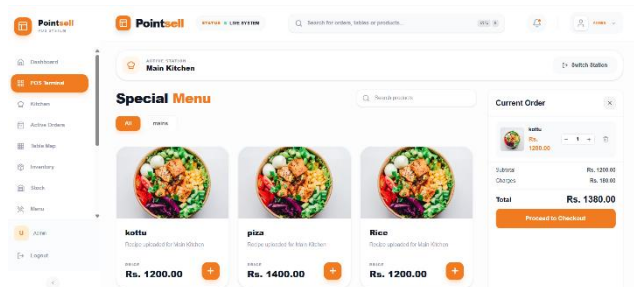


09/01/2026

Today, I focused on building the order-taking interface for waiters to streamline restaurant operations.

Progress Made:

- Designed order entry screen with:
  - Product category tabs
  - Visual product cards with images
  - Special instructions field
  - Order item list with modifications
- Implemented modifier selection:
  - Sizes, add-ons, cooking preferences, allergies/restrictions
- Added order routing options (kitchen/bar)
- Created order preview before submission
- Implemented course timing (appetizers, mains, desserts)
- Added rush order flag for priority handling



## SUPERVISOR COMMENTS FOR THE WEEK

*Gamage had a strong start to the month by proactively enhancing the POS system's security, implementing rate limiting, Helmet.js, and input sanitization to prevent attacks. She successfully set up the React frontend using Vite and efficiently managed complex states using React Context and the useReducer hook. Her UI/UX skills were highly evident in her development of the cashier dashboard, the drag-and-drop restaurant table management layout, and the detailed order-taking interface. She collaborated effectively with the team and utilized our progress evaluation meetings well to guide her designs.*

Supervisor's  
Signature



Date 3/4/2026

## Form I – 3A

## INTERN'S DAILY DIARY

(To be filled by the Intern- Please ensure to upload duly filled set of forms at end of four weeks to the provided folder)

## Intern's Information

Intern's Name

D.N Gamage

Student ID

IT23481244

Internship Title

Software Engineer (Intern)

Specialisation

Information Technology

Supervisor  
Name

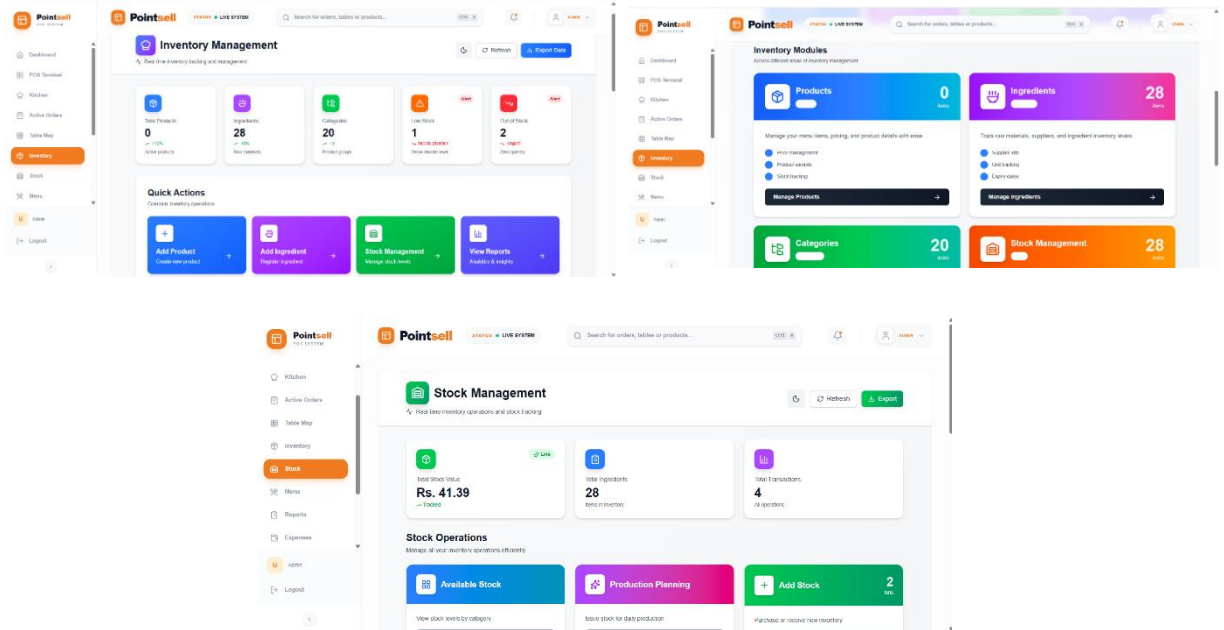
Bandara K.M.V.T

## Internship Information

## Training Information For the Week (to be filled by the intern)

| DATE       | DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., SKETCHES AND DIMENSIONS TO BE GIVEN WHEREVER POSSIBLE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12/01/2026 | <p>Today, I worked on building the Kitchen Display System interface to improve kitchen operations.</p> <p>Progress Made:</p> <ul style="list-style-type: none"> <li>Created dashboard showing:               <ul style="list-style-type: none"> <li>Active orders in queue</li> <li>Order age/timing indicators</li> <li>Item-level preparation status</li> <li>Alerts for delayed orders</li> </ul> </li> <li>Implemented order workflow: New → Acknowledged → Preparing → Ready → Served</li> <li>Added audio/visual alerts for new orders</li> <li>Created bump screen to mark items complete</li> <li>Separate views for kitchen and bar</li> <li>Order history view for reference</li> <li>Preparation time tracking for analytics</li> </ul>                                                                                                 |
| 13/01/2026 | <p>Today, I worked on implementing the inventory management interface, collaborating with H.M.L.P Abayarathne.</p> <ul style="list-style-type: none"> <li>Developed inventory dashboard showing:               <ul style="list-style-type: none"> <li>Current stock levels</li> <li>Low stock alerts</li> <li>Stock value summary</li> <li>Recent stock movements</li> </ul> </li> <li>Built stock management screens:               <ul style="list-style-type: none"> <li>Stock-in form with supplier details</li> <li>Stock-out form with reason tracking</li> <li>Stock adjustment with approval workflow</li> <li>Stock transfer between locations</li> </ul> </li> <li>Implemented inventory reports:               <ul style="list-style-type: none"> <li>Inventory valuation report</li> <li>Stock movement history</li> </ul> </li> </ul> |

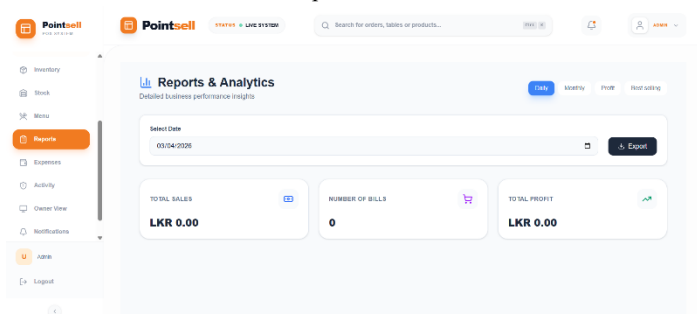
- Expiry date tracking
- Reorder point suggestions
- Added CSV import for bulk stock updates
- Created barcode printing for inventory labels



14/01/2026

Today, I focused on building the reporting and analytics dashboard.  
Progress Made:

- Developed main analytics dashboard with:
  - Daily sales summary cards
  - Sales trend charts (line/bar graphs)
  - Top-selling products
  - Revenue breakdown by payment method
  - Hourly sales heatmap
- Implemented interactive charts using Chart.js
- Built date range selector for all reports
- Added export functionality (CSV, PDF)
- Created print-optimized layouts
- Enabled real-time dashboard updates



15/01/2026

Thaipongal Holiday

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 16/01/2026 | <p>Today, I worked on the admin user management module.</p> <ul style="list-style-type: none"> <li>• Built user management screen with: <ul style="list-style-type: none"> <li>○ List of users with role badges</li> <li>○ Active/inactive toggle</li> <li>○ User creation and edit forms</li> <li>○ Password reset functionality</li> </ul> </li> <li>• Implemented role and permission management: <ul style="list-style-type: none"> <li>○ Assign roles to users</li> <li>○ Display permission matrix</li> <li>○ Create custom roles</li> </ul> </li> <li>• Added user activity log viewer</li> <li>• Developed staff performance metrics: <ul style="list-style-type: none"> <li>○ Sales per user</li> <li>○ Average transaction value</li> <li>○ Transaction count</li> <li>○ Login history</li> </ul> </li> <li>• Implemented session management for security</li> </ul> |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## SUPERVISOR COMMENTS FOR THE WEEK

*Gamage demonstrated excellent frontend capabilities this week by developing the Kitchen Display System (KDS) and a comprehensive inventory management interface with barcode printing capabilities. She successfully built an interactive reporting and analytics dashboard utilizing Chart.js, providing valuable business insights such as hourly sales heatmaps and revenue breakdowns. Additionally, she implemented a robust admin user management module featuring role-based access control, an activity log viewer, and staff performance metrics.*

Supervisor's  
Signature



Date 3/4/2026

## Form I – 3A

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Student ID

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Specialisation

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Supervisor  
Name

Bandara K.M.V.T

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|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 19/01/2026 | <p>Today, my supervisor conducted a board meeting with all team members to review project progress. I worked on the settings and configuration module.</p> <p>Progress Made:</p> <ul style="list-style-type: none"> <li>Developed settings screens for:               <ul style="list-style-type: none"> <li>General business info (name, address, contact)</li> <li>Tax/VAT configuration</li> <li>Receipt customization</li> <li>Payment method setup</li> <li>Printer settings</li> </ul> </li> <li>Implemented validation for:               <ul style="list-style-type: none"> <li>Required fields</li> <li>Format and range checks</li> </ul> </li> <li>Added settings preview before saving</li> <li>Implemented backup and restore for settings</li> <li>Created setup wizard for initial system configuration</li> </ul>                                                                                                                                               |
| 20/01/2026 | <p>Today, I focused on making the POS system fully responsive and optimized for mobile/tab devices.</p> <ul style="list-style-type: none"> <li>Applied mobile-first responsive design across all interfaces:               <ul style="list-style-type: none"> <li>Mobile screens for key operations</li> <li>Tablet layouts for POS and kitchen displays</li> <li>Desktop views for admin functions</li> </ul> </li> <li>Added mobile-specific enhancements:               <ul style="list-style-type: none"> <li>Touch gestures and simplified navigation</li> <li>Larger touch targets for ease of use</li> </ul> </li> <li>Configured Progressive Web App (PWA) features:               <ul style="list-style-type: none"> <li>Service worker for offline support</li> <li>App manifest for install prompt</li> <li>Device-specific icon sets</li> </ul> </li> <li>Tested on multiple devices and screen sizes</li> <li>Fixed 12 layout and responsiveness issues</li> </ul> |

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 21/01/2026 | <p>Today, I focused on setting up and running integration tests for the POS system APIs.</p> <p>Progress Made:</p> <ul style="list-style-type: none"> <li>Configured Jest as the testing framework.</li> <li>Created a test database with seed data for consistent testing.</li> <li>Wrote integration tests for key flows: <ul style="list-style-type: none"> <li>Authentication (registration/login)</li> <li>Product CRUD operations</li> <li>Sales transactions and inventory updates</li> <li>Payment processing (cash, card, mobile)</li> </ul> </li> <li>Implemented test fixtures and factories for reusable test data.</li> <li>Added test coverage reports to track code tested.</li> <li>Configured CI/CD pipeline to run tests automatically.</li> <li>Documented testing steps and best practices.</li> </ul> |
| 22/01/2026 | <p>Today, I focused on E2E testing to validate complete user workflows in the POS system.</p> <p>Progress Made:</p> <ul style="list-style-type: none"> <li>Configured Cypress testing framework.</li> <li>Created E2E test scenarios for critical flows: <ul style="list-style-type: none"> <li>User login and logout</li> <li>Retail sale process</li> <li>Restaurant order workflow</li> <li>Inventory stock-in operations</li> <li>Report generation and export</li> </ul> </li> <li>Set up test fixtures for consistent test data.</li> <li>Added visual regression testing to catch UI changes.</li> <li>Recorded test sessions for bug tracking and documentation.</li> <li>Documented all test scenarios and expected results.</li> </ul>                                                                           |
| 23/01/2026 | <p>Today, I worked on improving the database and managing schema changes for the POS system. I optimized the database by adding indexes and refining relationships between tables to make queries faster and more efficient. I also set up a migration system using Sequelize, which allows us to safely update the database structure, track changes, and roll back updates if needed. Seed data was added for testing, and old transactions were planned for archiving so they don't slow down the system. Additionally, I automated backups and documented the schema with ER diagrams to make it easier for the team to understand. This work ensures the database is fast, reliable, and easy to maintain as the system grows.</p>                                                                                    |

## SUPERVISOR COMMENTS FOR THE WEEK

*This week, Gamage focused heavily on system reliability and configuration, successfully building the settings module and a setup wizard. She effectively applied a mobile-first approach to make the POS system fully responsive and configured it as a Progressive Web App (PWA) with a service worker for offline support. Her dedication to software quality was outstanding, as she established comprehensive integration testing using Jest, end-to-end testing with Cypress, and automated test runs via a CI/CD pipeline. Furthermore, she demonstrated strong backend capabilities by optimizing the database with indexes and establishing a safe migration system using Sequelize.*

Supervisor's  
Signature



Date 3/4/2026



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|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 26/01/2026 | Today, I focused on improving error handling and logging for the POS system. I created custom error classes and consistent response formats so users receive clear, understandable messages when something goes wrong. I set up Winston for structured logging of errors, access, performance, and audit data, with log rotation to manage file sizes. I also integrated Sentry for real-time error tracking in production and added notifications for critical issues. These improvements help the team quickly identify and fix problems, making the system more reliable and easier to maintain.                                                                                                                                                                                                                                              |
| 27/01/2026 | <p>Today POS development team Focused on documentation and knowledge transfer for the POS system.</p> <p>Progress Made:</p> <ul style="list-style-type: none"> <li>Prepared detailed technical documentation:               <ul style="list-style-type: none"> <li>System architecture overview</li> <li>API documentation using Swagger/OpenAPI</li> <li>Database schema and table relationships</li> <li>Deployment and troubleshooting guides</li> </ul> </li> <li>Created user manuals for different roles:               <ul style="list-style-type: none"> <li>Cashiers, Waiters, Kitchen staff, Managers, Administrators</li> </ul> </li> <li>Produced video tutorials for common operations.</li> <li>Documented environment setup and installation procedures.</li> <li>Compiled a FAQ covering common issues and solutions.</li> </ul> |

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 28/01/2026 | <p>Worked on minor improvements and general system maintenance.</p> <ul style="list-style-type: none"> <li>• Fixed small UI issues reported by users.</li> <li>• Improved form validations in sales and inventory modules.</li> <li>• Cleaned up unused code and optimized components.</li> <li>• Reviewed system logs for errors and resolved minor bugs.</li> <li>• Updated project documentation with recent changes.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 29/01/2026 | <p>Today, I began working on the preparation of the User Manual for the Delmege ERP system used at Delmage Technologies – Broker Company.</p> <p>I carefully analyzed the entire system to understand its functionality, workflow, and target users such as administrators, brokers, insurance company staff, and client company users.</p> <p>I reviewed all major modules of the system, including:</p> <ul style="list-style-type: none"> <li>• Dashboard</li> <li>• RAQ Management</li> <li>• Sectors Management</li> <li>• Policy Management</li> <li>• Client Company Management</li> <li>• Insurance Company Management</li> <li>• User Management</li> <li>• NatCat Exposure Dashboard</li> <li>• Settings</li> <li>• Job Management</li> </ul> <p>After reviewing each module, I identified the key user operations and features that need to be clearly explained in the manual. I then created the structure of the user manual document, including:</p> <ol style="list-style-type: none"> <li>1. Introduction</li> <li>2. System Overview</li> <li>3. Module-by-Module Instructions</li> <li>4. Step-by-Step Operational Guidelines</li> <li>5. Troubleshooting &amp; Support</li> </ol> <p>By the end of the day, I finalized the document outline and prepared a structured template for professional documentation.</p> |
| 30/01/2026 | <p>I focused on writing detailed content for each module of the Delmege ERP system.</p> <p>I started with the Introduction section, explaining:</p> <ul style="list-style-type: none"> <li>• The purpose of the Delmege ERP system</li> <li>• How it supports broker operations</li> <li>• The importance of proper system usage</li> </ul> <p>Then I documented step-by-step instructions for:</p> <ul style="list-style-type: none"> <li>• Creating and managing RAQ periods</li> <li>• Managing Sectors and assigning companies</li> <li>• Creating, editing, and managing insurance policies</li> <li>• Adding and updating client companies</li> <li>• Registering insurance companies</li> <li>• Managing user roles and permissions</li> </ul> <p>For each module, I:</p> <ul style="list-style-type: none"> <li>• Described the dashboard layout</li> <li>• Explained summary cards and filters</li> <li>• Documented search and navigation features</li> <li>• Added step-by-step procedures for adding, editing, and deleting records</li> </ul> <p>I also included explanations for status indicators, role-based access control, and dashboard analytics.</p> <p>By the end of the day, all core modules were fully documented with clear and user-friendly explanations.</p>                                               |

### 1. Introduction

#### About Delmege ERP

Delmege ERP is a comprehensive Enterprise Resource Planning solution designed specifically for broker companies and insurance management operations. This system integrates multiple business processes into a unified platform, enabling efficient management of companies, policies, sectors, client relationships, and insurance operations.

#### Purpose of This Manual

This user manual provides step-by-step instructions for navigating and utilizing the Delmege ERP system. It covers all major features, from basic navigation to advanced module management, ensuring users can maximize the system's capabilities effectively.

#### System Overview

The Delmege ERP system includes the following core modules:

- Dashboard - Real-time metrics and quick navigation
- RAQ Management - Request for quotation handling
- Sectors Management - Industry classification and tracking
- Policies Management - Insurance policy administration
- Client Companies Management - Customer relationship management
- Insurance Companies Management - Insurance provider management
- User Management - System access and permissions
- NatCat Exposure - Natural catastrophe risk assessment
- Settings - System configuration
- Jobs - Task and workflow management

#### System Access

To access the Delmege ERP system:

- Open your web browser (Chrome, Firefox, Safari, or Edge recommended)
- Navigate to the system URL
- Enter your username and password provided by your system administrator
- Click the login button to access the dashboard

#### User Interface Overview

The Delmege ERP interface consists of several key components designed for intuitive navigation and efficient workflow management.

- Top Navigation Bar

- Notification bell icon
- User profile dropdown (delmegeadmin)

The sidebar provides quick access to all system modules with visual icons and labels for easy identification.

### 1.1 Dashboard

The Dashboard is your central hub for monitoring key business metrics and accessing frequently used functions. It provides real-time insights into company performance, policy status, and system activities.

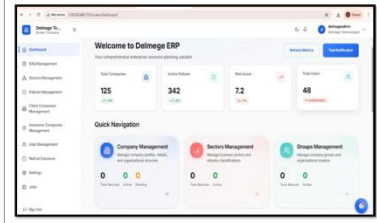


Figure 1-1 Delmege ERP Dashboard Overview

#### Key Metrics Cards

The top section of the dashboard displays four essential metrics that provide a quick snapshot of your business status:

- Total Companies
  - Current count: 125 companies
  - Growth indicator: +15% increase since ultimate previous month

## SUPERVISOR COMMENTS FOR THE WEEK

*Gamage successfully finalized her POS system tasks by implementing structured error handling with Winston and Sentry, alongside producing comprehensive technical documentation and Swagger API guides. Transitioning smoothly to a new assignment, she demonstrated strong analytical and technical writing skills by structuring and drafting the comprehensive User Manual for the Delmege ERP system. She effectively documented complex modules such as RAQ Management, Policy Management, and role-based access controls, ensuring clear, step-by-step guidance for all end-users.*

Supervisor's  
Signature

*Vihina*

Date 3/4/2026

## Form I – 3A

## INTERN'S DAILY DIARY

(To be filled by the Intern- Please ensure to upload duly filled set of forms at end of four weeks to the provided folder)

## Intern's Information

Intern's Name

D.N Gamage

Student ID

IT23481244

Internship Title

Software Engineer (Intern)

Specialisation

Information Technology

Supervisor  
Name

Bandara K.M.V.T

## Internship Information

## Training Information For the Week (to be filled by the intern)

| DATE       | DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., SKETCHES AND DIMENSIONS TO BE GIVEN WHEREVER POSSIBLE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 02/02/2026 | <p>I focused on reviewing and refining the entire user manual document.</p> <p>I performed the following tasks:</p> <ul style="list-style-type: none"> <li>• Checked grammar and technical accuracy</li> <li>• Improved clarity and simplified complex explanations</li> <li>• Ensured consistent formatting (headings, numbering, spacing)</li> <li>• Verified that all modules were properly covered</li> <li>• Organized screenshots with figure numbers and captions</li> <li>• Cross-checked navigation paths with the actual system</li> </ul> <p>I also added:</p> <ul style="list-style-type: none"> <li>• System Access instructions</li> <li>• Common Operations and Status Indicators</li> <li>• Built-in Guide explanation</li> <li>• Best Practices for system usage</li> <li>• FAQ and Support information</li> </ul> <p>Finally, I formatted the document professionally and prepared the final version in PDF format suitable for official submission at Delmage Technologies – Broker Company.</p> <p>The User Manual for the Delmege ERP / Broker Management System was successfully completed and finalized.</p> |

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| 03/02/2026 | <p>Today, Mr. K.M.V.T. Bandara assigned me to review the UniHub platform, which is currently under development. UniHub is a web platform built with Next.js and TypeScript that combines E-Learning and Marketplace functionalities.</p> <ul style="list-style-type: none"> <li>• Studied the overall system architecture and platform features.</li> <li>• Explored the E-Learning module including: <ul style="list-style-type: none"> <li>○ Course browsing and filtering</li> <li>○ Course enrollment process</li> <li>○ Learning dashboard and course progress tracking</li> <li>○ Course content access (video lectures and PDFs)</li> </ul> </li> <li>• Tested the student-side workflows to ensure navigation and feature accessibility.</li> <li>• Identified a few minor UI inconsistencies and loading delays in course listing pages.</li> <li>• Documented initial observations and prepared a checklist for further testing.</li> </ul>                                                                                         |
| 04/02/2026 | Holiday (Independence Day)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 05/02/2026 | <p>Today, I continued reviewing the UniHub platform, focusing mainly on the Tutor functionality in the E-Learning system.</p> <ul style="list-style-type: none"> <li>• Tested the Tutor Dashboard features including: <ul style="list-style-type: none"> <li>○ Course creation and editing</li> <li>○ Uploading course materials</li> <li>○ YouTube video integration for lectures</li> <li>○ Live session scheduling and management</li> <li>○ Student analytics and revenue tracking</li> </ul> </li> <li>• Verified form validations in the course creation process.</li> <li>• Checked course management features such as updating course details and pricing.</li> <li>• Identified a few issues: <ul style="list-style-type: none"> <li>○ Some form validation messages were unclear.</li> <li>○ Minor errors in live session scheduling workflow.</li> </ul> </li> <li>• Documented issues and possible improvements for the development team.</li> </ul>                                                                              |
| 06/02/2026 | <p>Today, I focused on testing the Marketplace module of the UniHub platform.</p> <ul style="list-style-type: none"> <li>• Tested buyer-side features: <ul style="list-style-type: none"> <li>○ Product browsing and search filters</li> <li>○ Product detail pages</li> <li>○ Shopping cart and checkout process</li> <li>○ Order tracking functionality</li> </ul> </li> <li>• Tested vendor-side features: <ul style="list-style-type: none"> <li>○ Vendor registration and shop setup</li> <li>○ Product listing creation with images</li> <li>○ Order management and shipping status updates</li> <li>○ Vendor dashboard analytics and performance metrics</li> </ul> </li> <li>• Identified several minor issues such as: <ul style="list-style-type: none"> <li>○ Some product images not loading correctly</li> <li>○ Minor UI misalignment in vendor dashboard</li> <li>○ Order status update delays</li> </ul> </li> <li>• Documented all findings and prepared a report on features that require fixes or improvements.</li> </ul> |

**SUPERVISOR COMMENTS FOR THE WEEK**

*Gamage began her final month by successfully completing and formatting the comprehensive User Manual for the Delmege ERP system, ensuring it was professional and ready for official submission. Transitioning to the new UniHub platform, she demonstrated strong analytical skills by thoroughly reviewing the existing E-Learning architecture. She effectively tested both student and tutor workflows—including course enrollment, YouTube integrations, and live session scheduling—proactively identifying and documenting UI inconsistencies and validation errors for the development team. Furthermore, her initial review of the Marketplace module's buyer and vendor features showed excellent attention to detail.*

**Supervisor's  
Signature****Date** 3/26/2026**Form I – 3A****INTERN'S DAILY DIARY**

*(To be filled by the Intern- Please ensure to upload duly filled set of forms at end of four weeks to the provided folder)*

**Intern's Information**Intern's Name **D.N Gamage**Student ID **IT23481244**Internship Title **Software Engineer (Intern)**Specialisation **Information Technology**Supervisor  
Name **Bandara K.M.V.T****Internship Information**

## Training Information For the Week (to be filled by the intern)

| DATE       | DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., SKETCHES AND DIMENSIONS TO BE GIVEN WHEREVER POSSIBLE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 09/02/2026 | <p>Today, I started working on the UniHub, Marketplace module with an initial discussion with Mr. Gayan Pathirage. The main focus was to understand the business goals, system scope, and how the Marketplace integrates with the existing UniHub platform.</p> <p>We identified the key user roles:</p> <ul style="list-style-type: none"> <li>• Buyers (students/users purchasing items)</li> <li>• Vendors (sellers listing products)</li> <li>• Admins (system management and monitoring)</li> </ul> <p>We discussed the expected user flow, including product browsing, purchasing, and vendor management. Special attention was given to secure transactions, user trust, and ease of use.</p> <p>Problems Identified:</p> <ul style="list-style-type: none"> <li>• Need for a smooth and simple user experience</li> <li>• Ensuring secure and reliable checkout process</li> </ul> <p>Solutions Discussed:</p> <ul style="list-style-type: none"> <li>• Clear UI/UX flow for buyers and vendors</li> <li>• Strong validation and secure payment handling</li> </ul> <p>Key Learning:<br/>Understanding business requirements clearly at the beginning helps reduce future rework and confusion.</p> |
| 10/02/2026 | <p>Today, I continued discussions with Mr. Gayan Pathirage to clarify both functional and non-functional requirements. We analyzed:</p> <ul style="list-style-type: none"> <li>• Performance expectations (handling multiple users)</li> <li>• Mobile responsiveness (important for student users)</li> <li>• Trust elements (ratings, seller info, product details)</li> </ul> <p>Problems Identified:</p> <ul style="list-style-type: none"> <li>• Image handling performance issues</li> <li>• Order status synchronization challenges</li> <li>• Complexity in integrating Marketplace with E-Learning module</li> </ul> <p>Solutions Proposed:</p> <ul style="list-style-type: none"> <li>• Optimize images using compression and lazy loading</li> <li>• Use proper backend state management for orders</li> <li>• Modular integration approach to avoid system conflicts</li> </ul> <p>Key Learning:<br/>Early identification of risks helps prepare better technical solutions.</p>                                                                                                                                                                                                                 |
| 11/02/2026 | <p>Today, I worked with the team to analyze the technical feasibility of the Marketplace module. The focus was on integrating Marketplace features with existing UniHub systems such as authentication, dashboards, and notifications.</p> <p>We reviewed:</p> <ul style="list-style-type: none"> <li>• System integration points</li> <li>• Role management for different users</li> <li>• Data handling and storage requirements</li> </ul> <p>Key Learning:<br/>Proper system planning is important to ensure smooth integration and scalability.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 12/02/2026 | <p>Today, I focused on designing the system architecture and database structure for the Marketplace module. We worked on:</p> <ul style="list-style-type: none"> <li>• Identifying key entities such as products, orders, vendors, and carts</li> <li>• Defining relationships between data tables</li> <li>• Planning scalable system architecture</li> </ul> <p><b>Key Learning:</b><br/>A well-structured database design improves system performance and data management.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

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| 13/02/2026 | <p>Today, I started implementing the Marketplace tab in the UniHub frontend system.</p> <p>Work completed:</p> <ul style="list-style-type: none"> <li>• Added Marketplace navigation</li> <li>• Created base layout and routing</li> <li>• Implemented role-based visibility</li> </ul> <p><b>Key Learning:</b><br/>A clear UI structure helps users navigate the system easily.</p> |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### SUPERVISOR COMMENTS FOR THE WEEK

*This week, Gamage showed excellent initiative in the planning and architectural phases of the UniHub Marketplace module. She actively participated in technical discussions to clarify business goals, identify key user roles, and define secure transaction flows. Her ability to anticipate non-functional requirements—such as image optimization for mobile responsiveness and order state management—was highly commendable. She successfully translated these requirements into a scalable database architecture and began the frontend implementation by establishing the base routing and role-based visibility for the Marketplace.*

Supervisor's  
Signature



Date 3/26/2026

## Form I – 3A

### INTERN'S DAILY DIARY

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#### Intern's Information

Intern's Name D.N Gamage

Student ID IT23481244

Internship Title Software Engineer (Intern)

Specialisation Information Technology

Supervisor  
Name Bandara K.M.V.T

#### Internship Information

### Training Information For the Week (to be filled by the intern)



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|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 16/02/2026 | <p>Today, I developed product browsing features for buyers. Before starting the implementation, I first designed the wireframes for the interfaces using Figma to plan the layout and user flow clearly.</p> <p>Features implemented:</p> <ul style="list-style-type: none"> <li>• Product listing display</li> <li>• Category filtering</li> <li>• Search functionality</li> </ul> <p><b>Key Learning:</b><br/>Planning UI with wireframes helps create better designs and improves user experience.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 17/02/2026 | <p>Today, I worked on developing the product detail pages and cart functionality for the Marketplace module. This involved creating a detailed view for each product where users can clearly see product images, descriptions, pricing, and availability before making a purchase decision. I also focused on improving the overall shopping experience by ensuring smooth interaction between product selection and the cart system.</p> <p>Work completed:</p> <ul style="list-style-type: none"> <li>• Designed and implemented product detail view with images, description, price, and stock information</li> <li>• Developed “Add to Cart” functionality with instant updates</li> <li>• Implemented cart features including add, remove, and quantity updates</li> <li>• Calculated and displayed subtotal dynamically based on selected items</li> </ul> <p><b>Key Learning:</b><br/>Real-time updates and smooth cart interactions improve usability and enhance overall user satisfaction.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 18/02/2026 | <p>Today, I focused on implementing the core functionality that allows buyers to finalize their purchases. The checkout process is a critical step in the user journey, as it directly affects conversion rates and overall user satisfaction.</p> <p>Features Added:</p> <ol style="list-style-type: none"> <li>1. Order Summary: <ul style="list-style-type: none"> <li>○ Developed a detailed order summary page displaying all items added to the cart.</li> <li>○ Included item name, quantity, price, subtotal, and total cost.</li> <li>○ Added functionality to update quantities directly within the summary for user convenience.</li> </ul> </li> <li>2. Address Input: <ul style="list-style-type: none"> <li>○ Implemented a form to collect shipping information, including full name, street address, city, postal code, and contact number.</li> <li>○ Added form validation to ensure required fields are correctly filled.</li> <li>○ Incorporated default address selection for returning users to speed up the checkout process.</li> </ul> </li> <li>3. Order Confirmation: <ul style="list-style-type: none"> <li>○ Built a confirmation page to display the final order details and shipping information.</li> <li>○ Integrated a confirmation button that triggers order placement and updates the database with order records.</li> <li>○ Added a notification mechanism to inform the user of successful order placement.</li> </ul> </li> </ol> |
| 19/02/2026 | <p>Today’s work concentrated on vendor onboarding. I implemented:</p> <ul style="list-style-type: none"> <li>• Registration flow – collecting vendor details with proper validation.</li> <li>• Profile setup – allowing vendors to create and update their shop profiles.</li> <li>• Basic shop initialization – enabling vendors to add initial information and products.</li> </ul> <p>Key Enhancements:</p> <ul style="list-style-type: none"> <li>• Introduced validation rules for required fields and proper formatting to ensure data integrity.</li> <li>• During a discussion with Mr. Gayan, we identified the need for a verification step before vendors become fully active.</li> <li>• Proposed a staged activation model: Registered → Under Review → Approved, reducing operational risk and aligning onboarding with platform quality standards.</li> </ul> <p>Schedule a meeting with Mr. K.M.V.T. Bandara to evaluate our progress and gather feedback on the onboarding workflow.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 20/02/2026 | <p>Today, I focused on building and refining the vendor-side product management features, which are critical for enabling vendors to list, manage, and maintain their inventory effectively. This functionality ensures that vendors have full control over their products, improving both platform usability and buyer experience.</p> <p>Features Developed:</p> <ol style="list-style-type: none"> <li>Product Creation: <ul style="list-style-type: none"> <li>Vendors can now add new products with comprehensive details, including: <ul style="list-style-type: none"> <li>Product name and description</li> <li>Pricing information, with validation for numeric values</li> <li>Category assignment for better organization and searchability</li> <li>Stock control, including quantity available and restocking options</li> <li>Multiple image uploads, allowing vendors to showcase products visually</li> </ul> </li> </ul> </li> <li>Product Editing: <ul style="list-style-type: none"> <li>Vendors can modify existing product details including images, pricing, categories, and stock levels.</li> <li>Implemented real-time updates so changes are immediately reflected in the vendor dashboard.</li> <li>Enabled image replacement and deletion for more flexible product management.</li> </ul> </li> <li>Validation and User Feedback: <ul style="list-style-type: none"> <li>Added form validation to ensure mandatory fields are correctly filled before submission.</li> <li>Implemented checks to ensure correct image formats and acceptable file sizes.</li> <li>Provided clear feedback messages to indicate success or failure of submissions.</li> </ul> </li> </ol> |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## SUPERVISOR COMMENTS FOR THE WEEK

*Gamage made rapid and highly effective progress on the Marketplace frontend this week. She utilized Figma to design clear wireframes before successfully implementing product browsing, category filtering, and search functionality. Her development of the product detail pages and the dynamic cart system showed a strong focus on the user experience. Furthermore, she successfully built the critical checkout workflow, including address validation and order confirmation. On the vendor side, she implemented a secure onboarding flow and comprehensive product management features, complete with multi-image uploads and real-time stock control.*

Supervisor's  
Signature



Date 3/26/2026

## Form I – 3A

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#### Intern's Information

Intern's Name

D.N Gamage

Student ID

IT23481244

Internship Title **Software Engineer (Intern)**

Specialisation **Information Technology**

Supervisor Name **Bandara K.M.V.T**

## Internship Information

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|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 23/02/2026 | <p>Today, the focus was on developing and refining vendor-side order management workflows. Efficient order handling is essential for timely fulfillment and maintaining buyer satisfaction, making this a critical component of the vendor dashboard.</p> <p>Features Implemented:</p> <ol style="list-style-type: none"> <li>Order List Views: <ul style="list-style-type: none"> <li>Developed a consolidated view of all orders for vendors, displaying key details such as order ID, buyer name, total amount, and current status.</li> <li>Implemented sorting and filtering options (e.g., by status, date) to improve usability.</li> </ul> </li> <li>Order Detail Inspection: <ul style="list-style-type: none"> <li>Created a detailed view for individual orders showing: <ul style="list-style-type: none"> <li>Product breakdown with quantities and prices</li> <li>Shipping and billing addresses</li> <li>Payment status and timestamps of key order events</li> </ul> </li> <li>Enabled vendors to verify and manage each order efficiently.</li> </ul> </li> <li>Shipping Status Updates: <ul style="list-style-type: none"> <li>Implemented workflow to update order statuses, transitioning orders logically through the lifecycle: Placed → Confirmed → Shipped → Delivered.</li> <li>Status updates trigger visual changes in both the vendor dashboard and buyer-facing interface.</li> </ul> </li> </ol> |
| 24/02/2026 | <p>In addition to Marketplace, I thoroughly tested Tutor functionality in the e-learning system to ensure parallel module stability. Covered features included course creation/editing, material uploads, YouTube lecture integration, live session setup, and analytics/revenue visibility. I verified form validation behavior and update flows for course details and pricing. Several validation messages appeared unclear to end users and were marked for rewording. This cross-module check ensured Marketplace additions were not degrading existing tutor workflows.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 25/02/2026 | <p>Today, I conducted broader QA scenarios covering both Marketplace and Tutor module workflows. The focus was to validate end-to-end user journeys and ensure stability across multiple modules after recent feature additions.</p> <p>Areas Tested:</p> <ol style="list-style-type: none"> <li>Marketplace Buyer &amp; Vendor Journeys: <ul style="list-style-type: none"> <li>Re-tested search, cart, checkout, and order tracking from the buyer perspective.</li> <li>Verified vendor-side product operations, including creation, editing, image uploads, stock management, and order handling.</li> </ul> </li> <li>Tutor Live-Session Workflows: <ul style="list-style-type: none"> <li>Validated scheduling, notifications, and update flows for live sessions.</li> <li>Checked for consistency in session creation, modification, and cancellation.</li> </ul> </li> </ol> <p>Observations &amp; Findings:</p> <ul style="list-style-type: none"> <li>Defects were reproduced and documented with clear steps, expected outcomes, and actual results.</li> <li>Key issues identified: <ul style="list-style-type: none"> <li>Minor UI misalignments in the vendor dashboard.</li> <li>Workflow inconsistencies in live session scheduling and notifications.</li> </ul> </li> </ul>                                                                                                                                  |

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 26/02/2026 | A focused triage session was conducted with Mr. Gayan to categorize and prioritize all identified issues. We grouped defects into immediate fixes, short-term improvements, and backlog enhancements. Immediate priorities included image reliability, clearer validation text, and order status responsiveness. For each item, I proposed actionable solutions and implementation direction to minimize regression risk. Ownership and sequencing were agreed, enabling the team to move quickly into corrective development. |
| 27/02/2026 | I began my boarding page contribution by discussing requirements with Mr. Gayan Pathirage and the development team. We defined the purpose of the feature: helping students and users find verified boarding places near universities. We identified key user needs such as location filtering, budget-based search, room details, and easy contact with property owners. I documented core functional requirements and clarified expected output for both user-facing pages and admin controls.                               |

## SUPERVISOR COMMENTS FOR THE WEEK

*Gamage demonstrated excellent full-stack capabilities and quality assurance skills this week. She successfully finalized the vendor-side order management workflows, allowing vendors to easily track and update shipping statuses. She then conducted rigorous cross-module testing to ensure her Marketplace additions did not degrade the existing Tutor functionalities, specifically checking live session scheduling and analytics. During a focused triage session, she effectively categorized and prioritized identified defects, proposing actionable solutions to minimize regression risks. She concluded the week by actively gathering requirements for a new student boarding feature.*

Supervisor's  
Signature



Date

3/26/2026

## Form I – 3A

## INTERN'S DAILY DIARY

(To be filled by the Intern- Please ensure to upload duly filled set of forms at end of four weeks to the provided folder)

## Intern's Information

Intern's Name

D.N Gamage

Student ID

IT23481244

Internship Title

Software Engineer (Intern)

Specialisation

Information Technology

Supervisor  
Name

Bandara K.M.V.T

## Internship Information

## Training Information For the Week (to be filled by the intern)

| DATE       | DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., SKETCHES AND DIMENSIONS TO BE GIVEN WHEREVER POSSIBLE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 02/03/2026 | I analyzed how the boarding feature should fit into the existing UniHub platform structure. I reviewed related modules such as authentication, navigation, and listing-style components already used in other sections. I prepared a task breakdown for UI design, backend integration, validation, and testing. I also listed possible risks like incomplete listing data, inconsistent image uploads, and user confusion in form submission steps.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 03/03/2026 | I focused on understanding the boarding-related database and APIs required for the page. I reviewed how fields like title, location, price, room type, facilities, contact details, and availability should be stored and returned. I checked response formats and identified mandatory vs optional data. This helped me design the page structure around real data behavior instead of only static assumptions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 04/03/2026 | <p>Today, I began the frontend implementation of the boarding page layout for the UniHub platform. The focus was on creating a visually consistent, responsive interface that presents boarding listings clearly and intuitively for students.</p> <p>Features Implemented:</p> <ol style="list-style-type: none"> <li>Listing Section:             <ul style="list-style-type: none"> <li>Developed a card-based preview layout for each boarding listing.</li> <li>Cards display key information, including:                 <ul style="list-style-type: none"> <li>Price per month or semester</li> <li>Location details for easy reference</li> <li>Quick summary details such as available rooms, facilities, or capacity</li> </ul> </li> <li>Implemented hover and click interactions for improved usability and navigation.</li> </ul> </li> <li>Responsive Design:             <ul style="list-style-type: none"> <li>Ensured the layout adapts seamlessly across desktop and mobile views.</li> <li>Tested spacing, text hierarchy, and visual hierarchy for readability and aesthetic consistency.</li> </ul> </li> <li>Sample Data Testing:             <ul style="list-style-type: none"> <li>Rendered multiple sample listings to validate card spacing, text alignment, and overall page flow.</li> </ul> </li> </ol> |

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            | <ul style="list-style-type: none"> <li>○ Verified that dynamic content (e.g., varying description lengths) displays correctly without breaking the layout.</li> </ul> <p>4. Design Alignment:</p> <ul style="list-style-type: none"> <li>○ Ensured that the boarding page adheres to the visual style and branding of UniHub, including colors, fonts, and component consistency.</li> <li>○ Incorporated the wireframe designs created in Figma as a reference for layout, card structure, and responsive breakpoints.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 05/03/2026 | <p>Today, I focused on enhancing the user experience for browsing boarding listings by implementing filtering and search functionality. These improvements allow students to quickly find accommodations that match their preferences while ensuring robust handling of edge cases.</p> <p>Features Implemented:</p> <ol style="list-style-type: none"> <li>Filters: <ul style="list-style-type: none"> <li>○ Added user-focused filtering options, including: <ul style="list-style-type: none"> <li>▪ Location – city, area, or proximity-based filtering</li> <li>▪ Price Range – minimum and maximum rent values</li> <li>▪ Boarding Type – single rooms, shared rooms, or full apartments</li> </ul> </li> <li>○ Designed filters to be intuitive and easy to adjust, supporting both desktop and mobile layouts.</li> </ul> </li> <li>Search Flow: <ul style="list-style-type: none"> <li>○ Implemented a search bar and query handling to allow users to quickly narrow down listings.</li> <li>○ Integrated search results with the filter state so selections remain consistent.</li> </ul> </li> <li>Validation and Edge Cases: <ul style="list-style-type: none"> <li>○ Tested filtering and search behavior for empty search results to ensure graceful handling.</li> <li>○ Verified behavior for incomplete or invalid filter values, preventing errors or broken layouts.</li> <li>○ Improved reset logic so users can easily return to the full listing view without manually clearing filters.</li> </ul> </li> <li>User Experience Enhancements: <ul style="list-style-type: none"> <li>○ Added real-time updates to the listing view as filters are applied.</li> <li>○ Ensured consistency with visual style and responsive layout, matching the overall UniHub design language.</li> </ul> </li> </ol> |
| 06/03/2026 | <p>I worked on the listing detail view to show complete boarding information. This included gallery images, detailed descriptions, amenities, rules, and owner contact options. I checked the fallback behavior for missing images or partial listing data. I improved user flow so detail navigation remains smooth from search results to the full listing view.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## SUPERVISOR COMMENTS FOR THE WEEK

*This week, Gamage showed strong system design skills by planning the integration of the new boarding feature into the existing UniHub platform. She carefully analyzed the required database fields and API response formats to ensure her frontend designs were data-driven. She successfully implemented the frontend layout, creating a responsive, card-based preview interface that seamlessly adapts across devices. Her focus on user experience was evident in her development of intuitive filtering options (location, price, boarding type) and the detailed listing views, which gracefully handle missing data.*

Supervisor's  
Signature



Date 3/26/2026

## Form I – 3A

### INTERN'S DAILY DIARY

*(To be filled by the Intern- Please ensure to upload duly filled set of forms at end of four weeks to the provided folder)*

#### Intern's Information

Intern's Name **D.N Gamage**

Student ID **IT23481244**

Internship Title **Software Engineer (Intern)**

Specialisation **Information Technology**

Supervisor  
Name **Bandara K.M.V.T**

#### Internship Information

#### Training Information For the Week (to be filled by the intern)

| DATE       | DETAILS AND NOTES OF WORK CARRIED OUT, PROBLEMS ENCOUNTERED AND HOW SOLVED ETC., SKETCHES AND DIMENSIONS TO BE GIVEN WHEREVER POSSIBLE.                                                                                                                                                                                                                                                       |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 09/03/2026 | I contributed to boarding submission/create listing functionality and focused on form reliability. I added and verified validations for required fields, numeric inputs, and contact information formatting. During testing, I noticed some validation messages were unclear, so I documented better wording suggestions. This helped improve usability for users adding new boarding places. |

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| 10/03/2026 | I integrated the page components with backend APIs and verified request/response behavior. I tested successful data retrieval, loading states, and failure scenarios. I added clearer handling for API errors so users receive understandable feedback instead of generic failures. I also ensured the page remains stable even when some listing fields are missing or delayed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 11/03/2026 | <p>I specifically tested listing image upload and display behavior. I verified how images were stored, retrieved, and presented across cards and detail pages. I found minor issues where some images did not load consistently, then documented probable causes and proposed fixes. I recommended fallback placeholders and improved validation at upload time.</p> <p>I performed full user journey testing from discovering a boarding place to viewing details and contacting owners. I checked transitions between listing page, detail page, and filter states. I validated pagination/scroll behavior and tested responsiveness on different screen sizes. I documented UX improvements to make the feature more intuitive for first-time users.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 12/03/2026 | Mid Examination (Academic Holiday)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 13/03/2026 | <p>On the last day of my internship, I focused on polishing the Boarding page UI and addressing previously identified issues to bring the feature closer to production-ready quality. This milestone marked the culmination of my contributions to the UniHub platform and provided an opportunity to consolidate my learning and development experience.</p> <p>Work Completed:</p> <ol style="list-style-type: none"> <li>UI Consistency and Refinement: <ul style="list-style-type: none"> <li>Improved spacing, alignment, and layout across all major sections of the boarding page.</li> <li>Enhanced label clarity and typography for better readability and intuitive navigation.</li> <li>Ensured that the design adheres consistently to the UniHub visual style, including responsiveness on both desktop and mobile.</li> </ul> </li> <li>Bug Retesting and Resolution Tracking: <ul style="list-style-type: none"> <li>Re-tested previously reported defects, confirming which issues were resolved and which remained pending.</li> <li>Documented remaining issues with clear steps and observations for future prioritization.</li> </ul> </li> <li>Final Touches and Quality Improvements: <ul style="list-style-type: none"> <li>Verified filtering, search, and listing interactions to ensure smooth user experience.</li> <li>Tested edge cases and reset logic for completeness.</li> <li>Made minor adjustments to cards, buttons, and interactive elements to enhance visual consistency and usability.</li> </ul> </li> </ol> |

## SUPERVISOR COMMENTS FOR THE WEEK

*In her final week, Gamage focused on system reliability and polish for the boarding module. She successfully integrated the frontend components with the backend APIs, implementing robust error handling and loading states. She conducted thorough end-to-end user journey testing, specifically focusing on image upload behaviors and form validations, documenting excellent UX improvements. On her final day, she polished the UI consistency across the platform and tracked the resolution of remaining defects. Gamage has been an outstanding intern, and her contributions to the UniHub platform have been highly valuable.*

Supervisor's  
Signature



Date 3/26/2026